Hybrid Adaptive Consensus: A Sovereign Finite-State-Machine Fallback for Modular Blockchains

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Abstract

Modular blockchains obtain settlement and data availability (DA) from independently operated layers. A chain whose block validity requires Bitcoin settlement receipts and DA receipts inherits the liveness of those layers: when either stalls, no proposal satisfies the validity predicate and the chain halts, although its validators are connected and correct. Dropping the requirement restores progress but suspends the guarantees it provided. We present *Engram Hybrid Adaptive Consensus*, a protocol layer on a CometBFT agreement core that makes this suspension an agreed, deterministic state transition. Engram adds (i) a *health certificate*, a deterministic aggregate of the height-bound, signed peripheral observations of a quorum, carried as vote extensions and bounded by the range of honest observations (Lemma 1), and (ii) a four-state finite state machine (ANCHORED, SUSPICIOUS, SOVEREIGN, RECOVERING) evaluated deterministically on that certificate. When Bitcoin, DA, or the peripheral P2P view degrades, the chain keeps producing blocks under the validator quorum, relaxes the receipt requirement of the unhealthy layer, and locks cross-chain withdrawals. It returns to ANCHORED after recovery hysteresis, DA backfill of the degraded interval, and a validator-verified zero-knowledge recovery proof (Noir/UltraHonk, 256 headers per proof).

A layered TLA+ specification, checked layer by layer with bounded model checking, has zero violations at $N \leq 4$ (the largest run explored 4.54 million distinct states); a four-mechanism ablation reproduces a distinct predicted failure for each removed mechanism. The cross-layer refinement obligation remains open. On a 7-validator Cosmos SDK testnet with live Bitcoin regtest and Celestia networks, a layered composite fault leaves the chain committing blocks through a full SOVEREIGN→RECOVERING→ANCHORED cycle with 100% AppHash agreement. The extended consensus object adds +228 bytes per proposal and 253 μs of validation per block. Against seven vanilla CometBFT validators on the same hosts, the block interval is 1.262 s against 1.233 s (about 2%; approximate 95% interval 0.3–4.4%), with every height committing in the first round. Against our hard-gated baseline, which enforces the same receipts at every height, Engram keeps 96–102% of its healthy block rate under a checkpoint pause, a DA outage and both, whereas the baseline falls to 56%, 16% and 17% and commits no block in the 120 s after the layer returns in all nine trials (n=3 per cell, one host). Engram's cost is the detection delay: 18–56 withdrawal-marked transactions were committed before it locked (41–117 s after the fault began). From the lock until the fault ended it committed none, which checks the circuit breaker against Engram's own transition rather than measuring safety independently; a baseline that keeps Engram's liveness but never locks lets 31–72 through in the same window.

Keywords: blockchain consensus, Byzantine fault tolerance, modular blockchain, finite state machine, formal verification, TLA+, zero-knowledge proofs

1. Introduction

Modular blockchains separate execution from services such as settlement and data availability (DA), which independently operated networks provide. Consider a chain whose validity predicate

requires fresh evidence from such layers: a block is valid only if it passes the BFT checks and carries both a Bitcoin settlement receipt and a receipt from a DA network such as Celestia [1], each attesting a height within a tolerance of the current Bitcoin tip (τ_{BTC}) or of the block (τ_{DA}). Formally,

$$valid(B) = BFTvalid(B) \wedge BTCreceipt(B) \wedge DAreceipt(B).$$

If Bitcoin settlement is congested or censored, or the DA network is unavailable, no proposal satisfies the predicate. The BFT consensus layer [2, 3] built on top stalls although the validator set is fully connected and correct. The chain inherits the liveness of the layers it depends on.

Dropping the requirement restores progress but suspends the guarantees that motivated the modular design: without a settlement checkpoint the chain cannot claim externally anchored settlement, and without DA confirmation it cannot claim that its recent data is externally available. The problem is therefore not only to detect an outage. The protocol must make all validators agree on when an external guarantee may be suspended, restrict the operations whose safety depended on it, and define the evidence required before the guarantee is restored.

The gap. Existing work addresses parts of this problem under other failure models. Protocols that trade finality for availability adapt to validator participation or consensus-network conditions, not to external infrastructure [4,5]. Rollup mechanisms provide alternative paths when sequencing or data publication fails, but the decision rests with users, a sequencer, a committee, or a contract on the parent chain [6–8]. Oracle designs and ABCI++ vote extensions let validators bring externally observed values into replicated state [9,10], but the value goes to the application. None of them covers a native chain whose validity depends on independently operated external layers that different validators observe differently. There, validators must agree on the observation and on the change in safety policy that follows from it (Section 2).

Why the decision belongs in consensus. The relaxation of receipt requirements and the withdrawal circuit breaker change two things that every validator must agree on: whether a block is valid, and whether a withdrawal executes. Bitcoin, DA, and peripheral-network status are observed locally and asynchronously, so two honest validators may see different conditions at the same instant. Suppose each validator applied a local policy to its own view. Validators would disagree on validity, so proposals would collect nil votes and rounds would repeat, or they would execute withdrawals differently and produce diverging `AppHash` values. A local, non-deterministic observation cannot drive replicated state directly. It must first become a deterministic input that every validator evaluates identically.

Design approach. Engram keeps CometBFT's proposer selection, prevote/precommit protocol, quorum intersection, and deterministic finality unchanged, and modifies the object and the validity policy that this pipeline carries. At height $h-1$, each validator signs its sensor snapshot as a vote extension; the sensors are read only there. The proposer of height h includes the signed snapshots of a quorum (more than two thirds of the voting power) in a *health certificate* C^h. Every validator verifies C^h, recomputes a deterministic weighted-median/majority aggregate $\mathsf{Agg}(C^h)$, and accepts a proposal only if its target state satisfies $s^h = \delta(s^{h-1}, \mathsf{Agg}(C^h))$, where δ is the transition function of a four-state finite-state machine (FSM) with states Anchored, Suspicious, Sovereign, and Recovering. FSM state is written only from a certificate that CometBFT has already agreed on (Section 4.3). "Hybrid Adaptive Consensus" denotes this adaptive layer inside the consensus pipeline, not a replacement for the agreement protocol.

The FSM makes the chain's security mode explicit. In Anchored, blocks require the Bitcoin and DA receipts. Sustained degradation moves the chain through Suspicious into Sovereign, where the quorum keeps ordering blocks, the receipt requirement of each unhealthy layer is relaxed, and cross-chain withdrawals are locked. Return to Anchored requires recovery hysteresis, DA backfill of the Sovereign interval, and a validator-verified zero-knowledge proof over that

interval (Sections 4.4 and 4.7; the prototype's proof gate is qualified in Section 8.3).

Scope. Engram handles degradation of Bitcoin, Celestia, and the peripheral P2P interfaces while the CometBFT quorum remains live. It does not address loss of the core quorum: like any BFT protocol, it needs at least a quorum (more than two thirds of the voting power) to communicate after global stabilization time. "P2P degradation" in this paper always means partial, peripheral degradation (Section 3.1).

To our knowledge, no published design combines the following for a modular chain whose validity depends on external settlement or DA layers: (i) a quorum-backed set of validator observations from which every validator deterministically derives external-layer health; (ii) deterministic adaptation of both block-validity requirements and withdrawal admissibility from that health while the native quorum keeps ordering blocks; and (iii) a recovery condition that requires a proof over the degraded interval and its DA backfill. Engram targets this combination.

Contributions.

- **Quorum-backed external-health agreement.** A height-bound certificate of signed observations from a quorum, carried in vote extensions, and a deterministic weighted-median/majority aggregation rule. When Byzantine validators hold less than one third of the voting power, every numeric aggregate stays within the range of the honest observations in a well-formed certificate, whichever observations a Byzantine proposer includes or omits (Section 4.3).
- **Dependency-aware adaptive execution.** A four-state FSM on the certificate, with asymmetric hysteresis and exponential backoff on repeated failed recovery, that changes the Bitcoin and DA receipt requirements without modifying CometBFT's agreement core. A withdrawal circuit breaker [11] locks cross-chain withdrawals while the chain runs without external corroboration (Sections 4.4–4.5; withdrawal-safety theorem in Section 5.1).
- **Proof- and backfill-gated recovery.** A Noir/UltraHonk circuit that proves header-chain continuity, FSM-transition legality, withdrawal-lock adherence, and boundary-validator signatures for up to 256 SOVEREIGN-interval headers at constant proving and verification cost; longer intervals chain several proofs (Appendix C). The exit condition also requires that the interval's data was published to the DA layer. Validators verify each proof inside block execution (Section 4.7).
- **Layered formal specification.** A four-layer TLA+ specification. Bounded model checking of each layer's safety and liveness properties found zero violations at $N \leq 4$; the largest single run explored 4.54 million distinct states. In a four-mechanism ablation study, removing each mechanism reproduces a distinct predicted counterexample. The cross-layer refinement obligation is specified but open (Section 5).
- **Live evaluation.** A 7-validator Cosmos SDK testnet against live Bitcoin regtest and Celestia networks with network- and application-level fault injection, comparing Engram against a hard-gated modular baseline and a vanilla CometBFT control (Sections 6–7).

The remainder of this paper is organized as follows. Section 2 situates Engram in the literature. Section 3 states the system model, the threat model, and the security goals. Section 4 defines the protocol: sensors, the health certificate, the FSM, proposal validity, and recovery. Section 5 states the security properties and reports the formal specification and bounded model checking. Section 6 describes the implementation, testbed, baselines, and methodology, and Section 7 answers RQ1–RQ4. Section 8 discusses the security envelope, limitations, and future work, and Section 9 concludes.

Table 1: Research Questions

RQ	Question
RQ1	**Safety.** Under Byzantine voting power below one third (two of the seven testbed validators), does certifying peripheral health by quorum prevent conflicting FSM states, forged or stale receipts, false health claims, and withdrawal leakage?
RQ2	**Liveness.** When Bitcoin or DA becomes unavailable, or the peripheral P2P view degrades while core quorum connectivity holds, does Engram keep committing blocks where a hard-gated modular baseline stalls?
RQ3	**Recovery.** Once peripherals recover, do hysteresis and the recovery proof return the chain to a stable ANCHORED state without oscillation?
RQ4	**Cost.** What overhead do the extended consensus object, certificate validation, and recovery proof add, and where does it come from: proposal size, validation time, or additional consensus rounds?

2. Related Work

A chain whose block validity depends on external layers raises five questions, and each has been studied by a separate community. Fig. 1 organizes the literature by these questions: how consensus behaves under degraded conditions (Section 2.1), how local observations of an external layer become data that all validators see and agree on (Section 2.2), how modular designs cope with the failure of Bitcoin, DA, or settlement layers (Section 2.3), how a chain returns to the anchored state (Section 2.4), and how such designs are validated (Section 2.5). Several building blocks of Engram exist in prior work: vote extensions that carry signed external data, weighted-median aggregation, and protocol-defined fallback for DA and sequencing failures. Engram differs in what the aggregated observation controls. Table 2 compares representative systems by trigger, decision authority, behavior under failure, recovery, assumption, and reported assurance; Section 2.6 states the combination left open.

2.1. Consensus Under Degraded Conditions

Agreement core. PBFT [50] and the partial-synchrony model of Dwork, Lynch, and Stockmeyer [51] underlie Tendermint and its CometBFT successor [2, 3], which provide deterministic finality with a rotating proposer. HotStuff [52] achieves responsiveness and linear communication under a correct leader within the same leader-based lineage. Narwhal/Tusk [53] separate transaction dissemination from ordering, and Bullshark [54] orders transactions on top of the Narwhal DAG. These protocols agree on the content of a proposal; whether a proposal is acceptable is decided by an application predicate that every correct validator must evaluate identically. Tendermint's consensus problem includes such a `valid()` predicate [3], and Cachin et al. [55] introduced the *external validity* condition for multi-valued Byzantine agreement. Engram relies on the same structure: its proposal-validation predicate depends on the health certificate, so the predicate stays deterministic only if the certificate is itself agreed on. BFT forensics [56] studies how many malicious replicas can be identified after a safety violation. AdoB [57] mechanizes benign and Byzantine consensus in one model, and Adore [58] adds certified reconfiguration. Engram inherits CometBFT's evidence pool for accountability and leaves the agreement core unchanged.

Fast paths and protocol switching. A second group changes the protocol's behavior in response to conditions that the replicas themselves observe. Abstract [13] lets a replicated state machine abort a client request when something goes wrong, which supports composing and reconfiguring BFT protocols. Thunderella [12] combines an optimistic fast path with a synchronous fallback, and Autobahn [14] targets seamless recovery from network blips. The trigger in each case is a fault of the replicas or of the network among them.

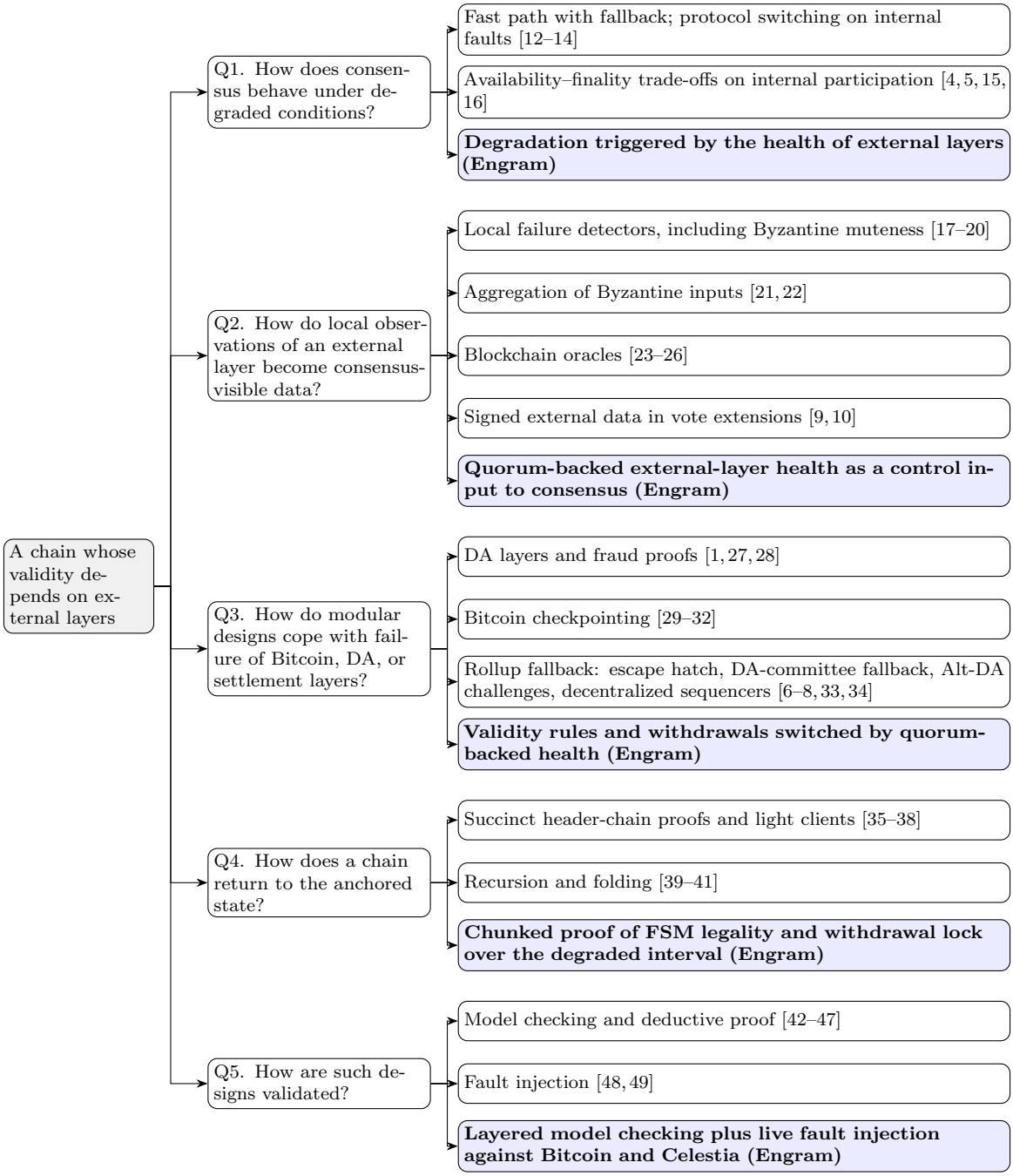


Figure 1: Taxonomy of related work, organized by the question each line of work answers. Shaded leaves mark where Engram sits. Vote-extension oracles supply signed external data to consensus (Q2) but attach no fallback semantics to it.

Availability–finality trade-offs. The closest conceptual relatives give up finality to keep the chain growing. The sleepy model [15] lets players be online or offline and gives a consensus protocol that is resilient as long as a majority of the online players are honest. Ebb-and-flow protocols [4] expose both a dynamically available ledger and a finalized prefix, so the chain keeps growing when finality stalls and the finalized ledger catches up on recovery. Casper FFG [16], and Gasper [5], which combines it with the LMD GHOST fork-choice rule, prove accountable safety and plausible liveness. For the case in which more than a third of the validators crash, Casper FFG proposes an inactivity leak that drains the deposits of non-voting validators until the voting validators again form a supermajority, and leaves the exact formula open. These designs share the "degrade, do not halt" goal and the tension described by the CAP theorem [59]. They differ from Engram in two respects: the trigger is *internal* participation, and the response changes how finality is reached, not which blocks and withdrawals are valid.

2.2. From Local Observations to Consensus-Visible External Data

Failure detectors. Unreliable failure detectors [17], their QoS metrics [18], and accrual detectors [19] give each process a local, eventually accurate suspicion of other processes. Muteness failure detectors [20] extend the abstraction to Byzantine processes that stop sending the messages an algorithm expects while possibly continuing to send others. Engram's sensors play the same role for peripheral layers, with two differences: a local suspicion has no effect until a quorum of signed observations backs it, and the adversary controls up to f of the reporters.

Aggregating Byzantine observations. Approximate agreement [21] requires non-faulty processes to decide on values that are close together rather than identical. Median validity [22] requires the decision to be close to the median of the non-Byzantine inputs and, at maximal fault tolerance with $n \geq 3t + 1$, within the range of those inputs. The Engram health certificate satisfies an analogous range-validity property for each weighted-median metric. It does not run a second approximate-agreement protocol: validators sign individual height-bound sensor snapshots carried as vote extensions, CometBFT agrees on the proposal that carries them, and every validator then recomputes the aggregate deterministically (Section 4.3).

Blockchain oracles. Oracle research addresses how replicated applications import external information. Town Crier authenticates web data with trusted hardware [23]. A systematization of oracle designs classifies their trust assumptions and manipulation risks [24]. DecenTruth combines multi-source truth discovery with decentralized consensus to tolerate dishonest reports [25], and PSCBO analyzes attacks on consensus-based oracle designs [26]. In all of these, the oracle supplies a value to the application layer.

Vote extensions. The closest mechanism to Engram's certificate is the ABCI++ vote extension. Each validator extends its precommit with application data, which may be obtained non-deterministically, and the next proposer includes the signed extensions in its proposal. The Cosmos SDK design record requires signatures from at least two thirds of the voting power and uses an oracle price fetcher as its example [9]; Slinky applies the pattern to off-chain price data and aggregates the extensions with a stake-weighted median [10]. Engram uses the same transport: sensor snapshots taken at height H reach the proposer at $H+1$, where their deterministic aggregation forms the health certificate. The difference lies in what the aggregate controls. A price oracle supplies a value to the application; the health certificate describes the infrastructure on which block validity depends and determines the receipt requirements, the withdrawal policy, and the entry into and exit from a degraded mode.

Peripheral P2P attacks. Sybil attacks [60], in which one entity presents multiple identities, and eclipse attacks on Bitcoin's P2P layer [61, 62], which monopolize a node's connections, let an adversary control what a node sees of an external network. They motivate the multi-signal

peer-health profile that Engram's P2P sensor uses and are one reason a single local reading is not trusted.

2.3. Fallback for Modular-Chain Dependencies

Data availability. LazyLedger/Celestia [1] decouples DA from execution. Fraud and data availability proofs [27] let light clients detect withheld data, and coded distributed arrays target the peer-to-peer layer that distributes data availability sampling fragments [28]. This line of work makes DA *verifiable*, but a chain that requires a DA inclusion proof before accepting a block still stops when DA stops. Engram adds a consensus-level response to DA unavailability and a backfill requirement before the chain returns to ANCHORED (Section 4.7).

Bitcoin anchoring. Pegged sidechains [63] introduced two-way pegs to Bitcoin. Karakostas and Kiayias [29] study checkpointing as a defense of proof-of-work ledgers against 51% attacks and report a liveness attack, "block lead", against existing checkpointing designs. For proof-of-stake chains, Babylon [30, 32] and Pikachu [31] checkpoint onto Bitcoin to prevent long-range attacks. Tas et al. [32] show that slashable safety is impossible without an external source of trust and state their guarantees under the assumption that Bitcoin is safe and live. Their design also defines a fallback in which Bitcoin orders the PoS blocks after censorship, which strengthens the PoS chain's dependence on Bitcoin. To our knowledge, none of these publications specifies how the chain operates while the anchor itself is unreachable. Engram treats unreachability as an input backed by a quorum of signed observations, keeps ordering blocks, and locks withdrawals while it persists. Accountability comes from CometBFT's equivocation evidence; Engram implements no Bitcoin-enforced slashing.

Rollup fallback and its decision locus. When a rollup operator stops building blocks or proposing state roots, escape hatches [6] let users withdraw on the L1 by proving their assets against an L2 state root. Arbitrum [64] resolves disputes about contract behavior with a challenge-based protocol on the L1. Surveys of layer-two protocols [65] treat the ability of honest users to withdraw as a core security property, and list among the open weaknesses that layer-two designs assume the underlying blockchain eventually accepts transactions. Capretto et al. [33] give fraud-proof games, mechanized in Lean 4, with which an honest agent can prove on the L1 that a sequencer-and-DAC "arranger" violated a safety property even if all of its replicas are compromised; a liveness violation that preserves safety cannot be proved on the L1. A follow-up decentralizes the sequencer and the DA committee with set consensus [34].

Two designs define a protocol-level response to a DA failure. AnyTrust normally relies on a data-availability committee. If the sequencer cannot collect enough committee signatures within a few minutes, it posts the full data to the parent chain as a non-AnyTrust rollup would do [7]. Withdrawals then incur the delay of a standard rollup, and the chain switches back to the cheaper mode when the committee resumes [66]. The sequencer makes this decision, and the failure it reacts to is a committee failure. OP Stack Alt-DA keeps data with an external DA provider and lets anyone challenge a commitment on the L1. If the data is not posted before the resolve window elapses, derivation reorgs the L2 blocks that depended on it [8]. This design answers the same question as Engram, what happens when an external DA input fails, by resolving the failure on the L1 and discarding the affected blocks.

These designs show that protocol-defined fallback for DA and sequencing failures exists. They differ from Engram in three respects:

- *Who decides.* A sequencer, a committee, users, or an L1 contract; Engram decides inside the validator set's own consensus instance.
- *What is observed.* A committee or data-posting failure, not the health of Bitcoin settlement or of an external DA network; Engram uses signed observations of all three peripheral

layers.

- *What the response changes.* It publishes more data or reorganizes the chain; Engram changes the validity predicate and the withdrawal policy and commits the FSM state in the same block.

Withdrawal risk. Lee et al. [67] break a bridge into its components, identify the risk each introduces, and analyze past exploits that targeted bridges. They report that over USD 1 billion in digital assets was stolen, incorrectly minted, or locked in bridges in the year before their study. Withdrawals are the path through which a chain without external corroboration can lose assets, which is why Engram locks them while the chain runs in SOVEREIGN mode.

2.4. Recovery and Re-Anchoring

Succinct proofs are the standard tool for convincing a verifier of a chain's history without re-executing it. They differ in how a long history is covered. Mina [37] keeps a constant-size proof of the whole chain by recursively composing SNARKs, which Plumo's authors report imposes a large overhead on the protocol being proved [36]. zkBridge [35] verifies source-chain consensus inside a two-layer recursive proof system. Plumo instead syncs a client through a chain of sequential SNARK transition proofs, each covering four months of blockchain history [36]. Recursive composition [39, 40] and folding [41] are the general techniques behind the recursive designs. FlyClient [38] takes a different route for proof-of-work chains: a light client downloads only a logarithmic number of block headers, selected by probabilistic block sampling. All of these prove that a header chain, or a validator-set transition, is valid. A chain that has run without external corroboration must also show that the interval followed the protocol. The long-range attack literature [68] reviews how an attacker can rewrite the history of a proof-of-stake chain, which is the exposure of an interval secured only by the chain's own validators. Engram's recovery proof is a fixed-capacity PLONK-family [69] proof over one interval chunk, and chunk proofs are chained through an on-chain root, as Plumo chains its transition proofs. The statement differs: it covers FSM-transition legality and withdrawal-lock adherence in addition to header continuity (Section 4.7).

2.5. Validation Methodology

TLA+ [70] with TLC and Apalache [42] has been used to specify and model-check Tendermint's blockchain synchronization protocol [43]. Deductive approaches establish properties by machine-checked proof instead of by exploring a bounded instance. Ivy [44] verifies safety of infinite-state systems interactively, and Lotan and Shoham [71] automate liveness proofs in first-order logic over finite but unbounded domains. IronFleet [45] and Verdi [46] verify distributed-system implementations, the latter in Coq, and LiDO [47] gives mechanized safety and liveness proofs in Coq for unpipelined and pipelined Jolteon. A recent preprint places these approaches on a maturity scale [72]. Engram sits in the model-checking family, with a per-layer result and an open refinement obligation (Section 5). On the empirical side, Twins [48] tests BFT implementations by duplicating validator identities, and chaos engineering [49] injects faults into production systems. Engram's evaluation combines network-level and application-level fault injection against live Bitcoin and Celestia networks.

2.6. Research Gap

Prior work supplies consensus that adapts to internal participation faults, oracle systems that aggregate external observations, vote extensions that deliver signed external data to the validator pipeline, rollup fallbacks for sequencing and DA failures, and succinct proofs of chain histories. When an external layer fails, a chain that gates validity on it halts, and a chain that drops the

dependency gives up the guarantees that motivated it. The fallbacks surveyed above place the decision with users, watchers, a sequencer, a committee, or an L1 contract.

To the best of our knowledge, no published design combines (i) a quorum-backed set of height-bound validator observations from which every validator deterministically derives the health of external layers, (ii) use of that health inside the validator consensus pipeline to change both proposal-validity requirements and withdrawal admissibility while the native validator quorum keeps ordering blocks, and (iii) a recovery condition that requires a proof over the degraded interval and the backfill of that interval to the DA layer before the chain returns to the externally anchored mode. Engram provides these three (with the proof-gate limitation of Section 8.3), retains CometBFT as the agreement protocol, and Sections 3–7 evaluate the combination.

3. System Model and Security Goals

This section fixes who participates, what may fail, what is trusted, and which properties the protocol is meant to give in each FSM state.

3.1. Validators, Network, and Assumptions

Validators and network. We assume partial synchrony: message delay is unbounded before an unknown global stabilization time (GST) and bounded after it [51]. Let V be the validator set, $w_i > 0$ the voting power of $i \in V$, $W = \sum_{i \in V} w_i$ the total voting power, and $\mathrm{wt}(S) = \sum_{i \in S} w_i$ the weight of $S \subseteq V$ (Table 3). The adversary controls a set $B \subseteq V$ of Byzantine validators, which may deviate arbitrarily and collude; the other validators $H = V \setminus B$ are honest. As in CometBFT, a *quorum* is a set whose weight exceeds two thirds of the total:

$$\mathrm{wt}(B) < \frac{W}{3}, \qquad \mathrm{wt}(S) > \frac{2W}{3} \ \text{ for every quorum } S. \tag{1}$$

The seven-validator testbed gives every validator the same power, so Eq. (1) allows f=2 Byzantine validators and a quorum is any five validators. The adversary cannot forge signatures, find hash collisions, or subvert the UltraHonk backend's KZG setup. After GST, the *core consensus network* connects a quorum, as CometBFT itself requires.

Validator set. The validator set is the genesis set and stays fixed throughout a SOVEREIGN or RECOVERING interval. Certificate signatures verify against the genesis validator keys, and the compiled recovery circuit fixes the set size at seven. Validator-set changes inside a degraded interval would require proving the set transitions and are outside this paper.

Peripheral layers. Bitcoin, Celestia, and the peripheral P2P interfaces may be slow, unavailable, partitioned from some validators, or adversarially degraded (delayed, censored, eclipsed) for arbitrarily long. They are not assumed to be Byzantine-fault-tolerant from the validator set's perspective, but Bitcoin reorganizations deeper than a configurable depth $K_{\text{DEEP_FINALITY}}$ are excluded, as in every Bitcoin-anchoring design (Section 2). Honest validators' peripheral observations may differ at any instant, and identical views are not assumed. The health certificate (Section 4.3) gives every validator the same aggregate under arbitrary honest disagreement; whether a sensor reading matches the external layer is a separate question (Section 4.2).

Parameters. FSM thresholds, hysteresis limits, and $K_{\text{DEEP_FINALITY}}$ are genesis parameters shared by every validator, like the block time or quorum size. A validator running different parameters simply fails to validate honest proposals; this affects only its own participation, not safety.

Layered adversary. The adversary may degrade the peripheral connectivity of *every* honest validator at once (delay Bitcoin headers, cut the DA bridge, eclipse the peripheral P2P interface),

Table 2: Failure handling in representative systems, from each system's primary publication or specification; "n/a": not stated by the source. The assurance column does not rank the designs.

System	Trigger and observation	Decided by	Behavior under failure and recovery	Assumption	Assurance reported
Reference point					
Hard-gated chain[a]	Missing Bitcoin or DA receipt	Validity predicate	Chain halts until the receipt returns	External layers live	n/a
Degradation inside consensus					
Ebb-and-flow [4]	Dynamic participation or network partition	Validator protocol	Ledger keeps growing and the finalized prefix lags; the prefix catches up	Partial synchrony (finalized ledger); synchrony (available ledger)[b]	Security proof; simulations
Casper FFG leak [16]	More than one third of validators stop voting	Validator protocol	Inactivity leak drains non-voting deposits; formula left open	PoW overlay; under 1/3 slashable stake (safety); 2/3 voting (liveness)	Accountable-safety and plausible-liveness proofs
External data in consensus					
Vote-extension oracle [9, 10]	External data update; signed vote extensions	Validator consensus pipeline	Stake-weighted median goes to the application; fallback not specified	Signatures from at least two thirds of voting power	Design record; open-source code
Bitcoin anchoring					
Babylon [32]	Transaction censorship; liveness transactions	Clients and Bitcoin[c]	Bitcoin orders PoS block hashes (rollup mode); the chain later exits rollup mode	Bitcoin safe and live	Security proofs; measured Bitcoin checkpoint latency
Rollup fallback					
Escape hatch [6]	No new state root within a time limit	Users	Users bypass the bridge and withdraw with a Merkle proof; a nullifier blocks a second escape	L1 live; users submit proofs	Design and use cases
AnyTrust [7, 66]	Too few committee signatures	Sequencer	Full batch posted to the L1 with standard rollup delay; the committee path resumes	L1 live	Public documentation
OP Alt-DA [8]	External DA input unavailable; L1 challenge	L1 contract	Data resolved on the L1, or affected L2 blocks reorged	L1 live; some party challenges	Experimental specification
Secure arranger [33]	Safety violation by the arranger	Honest agent and L1 contract	Fraud proof settles on the L1; liveness violations cannot be proved there	L1 safe and live; one honest agent	Fraud-proof games mechanized in Lean 4
Engram	Bitcoin, DA, or peripheral-P2P health; signed validator observations	Validator BFT pipeline	Receipt rules relaxed for the failed layer and withdrawals locked; proof, DA backfill, and hysteresis restore[d]	CometBFT quorum connected; Byzantine power under 1/3	Per-layer bounded model checking; live fault injection (one host)

Notes. (a) Engram's binary with the FSM pinned to ANCHORED (Section 6.4). (b) The finalized ledger tolerates under 1/3 adversarial nodes, the available ledger under 1/2. (c) Clients force the switch to rollup mode and apply the emergency break. (d) The prototype enforces DA backfill; a heuristic can satisfy the proof condition unless proof-only recovery is on; the refinement is open (Sections 8.3 and 5).

Table 3: Notation used in Sections 3–5.

Symbol	Meaning
V, w_i, W	validator set, voting power of validator i, total voting power
$\mathrm{wt}(S)$	total voting power of the validators in S
B, H	Byzantine and honest validators
h, r	consensus height and round
m_i, o_i	local metric vector and signed observation of validator i
C^h, Agg	health certificate of height h, deterministic aggregation
s^h, δ	FSM state committed at height h, transition function
τ_{BTC}, τ_{DA}	receipt tolerance of the Bitcoin receipt (Bitcoin blocks) and of the DA receipt (Engram blocks)

but it cannot partition the core consensus network below a quorum. Degrading the periphery changes the FSM state; it does not by itself stop block production.

Byzantine power of one third or more. Beyond $W/3$, no BFT protocol retains agreement. In Anchored and Suspicious, a checkpoint that carries the CometBFT block hash lets any Bitcoin observer detect a fork, because two conflicting histories cannot both match a Bitcoin-timestamped hash; this is accountability, not prevention. In Sovereign nothing outside the validator set constrains the chain, which is why withdrawals are locked in that state (Section 3.2).

3.2. Security Goals by Mode

We state what each FSM state guarantees, using four properties of the committed chain $B_1, \ldots, B_h$. A *prefix* $B_1, \ldots, B_c$ is the chain up to height $c \le h$.

- P_{agree}: no two conflicting blocks are committed at the same height. Assumption: $\mathrm{wt}(B) < W/3$.
- $P_{\mathrm{settle}}(c)$: the prefix up to c cannot be rewritten without a Bitcoin reorganization deeper than $K_{\mathrm{DEEP_FINALITY}}$. Assumptions: such reorganizations are excluded (Section 3.1), and the Bitcoin checkpoint that buries height c commits the chain through c (assumption M2 of Theorem 1, Section 5.1).
- $P_{\mathrm{avail}}(c)$: the block data of every height up to c is retrievable from the DA layer. Assumption: the DA layer is honest and the data was published.
- P_{exit}: a withdrawal commits only against a chain whose settled and available prefix is the one the withdrawal depends on, up to the receipt tolerance defined below. It is enforced by the circuit breaker.

Table 4 lists which properties hold in each state.

Table 4: Properties by FSM state. P_{agree} holds in every state under $\mathrm{wt}(B) < W/3$. † holds only if the checkpoint commits the chain through its height and the recovery exit gate includes DA backfill (Section 4.7). The prototype implements both; the proof condition of the exit gate is qualified in Section 8.3.

State	P_{agree}	P_{settle}	P_{avail}	Withdrawals
Anchored	Yes	Yes†	Yes†	Permitted
Suspicious	Yes	Yes†, delayed	Yes†	Permitted
Sovereign	Yes	No	No (validator block stores only)	Locked
Recovering	Yes	Pending recovery proof	Backfilling	Locked

Receipt tolerance. The two receipts are measured on different clocks, so each has its own tolerance, fixed by the round in which the block commits (Section 4.6). τ_{BTC} bounds, in Bitcoin

blocks, how far the checkpoint's Bitcoin height may lag the certified Bitcoin tip b_h. τ_{DA} bounds, in Engram blocks, how far the DA receipt's height may lag h. In ANCHORED and SUSPICIOUS, a valid block at height h whose layers are reported healthy carries a Bitcoin receipt that settles Engram height c_s through a checkpoint at Bitcoin height at least $b_h - \tau_{\mathrm{BTC}}$, and a DA receipt that attests availability up to Engram height $c_a \geq h - \tau_{\mathrm{DA}}$.

Theorem 1 (Section 5.1) states the conditions under which a committed withdrawal satisfies P_{exit}. Section 4 defines the mechanisms, and Section 5 states and checks the properties.

4. Engram Protocol

The mechanisms follow the order of the data: the architecture, the sensors that produce observations, the certificate that aggregates them, the state machine that consumes the aggregate, the validity predicate that enforces it, and the recovery path out of SOVEREIGN.

4.1. Architecture Overview

Engram is a Cosmos SDK application [73] running on a CometBFT core. Three peripheral layers are monitored: (i) Bitcoin, used as a settlement and finality anchor via periodic checkpoints; (ii) Celestia [1], used as the DA layer; and (iii) the peer-to-peer interfaces through which validators reach the Engram, Bitcoin, and Celestia networks. Engram follows one rule throughout: *sensors observe, the quorum certifies, the state machine decides.* Each validator's sensors read the peripheral layers locally and asynchronously, so two honest validators may observe different values at the same instant. Those readings never enter replicated state directly. Instead, each validator signs its reading as a CometBFT vote extension; the next proposer aggregates a quorum of them into a *health certificate* (Section 4.3); and the FSM transition is a deterministic function of that certificate and the previously committed FSM state (Fig. 2).

4.2. Network Sensors

Sensors are the only non-deterministic component; their outputs enter consensus only through the certificate (Section 4.3).

Bitcoin finality gap. $\Delta H_{\mathrm{BTC}} = H_{\mathrm{current}} - \min(H_{\mathrm{submitted}}, H_{\mathrm{anchored}})$ measures how far the latest checkpoint lags Bitcoin's tip; it grows under congestion or checkpoint censorship. Because a delay metric cannot detect a checkpoint that was reorganized away, each validator also re-verifies, against its own Bitcoin full node, that the checkpoint transaction is still included at depth $\geq K_{\mathrm{DEEP_FINALITY}}$. A failed re-verification is a critical condition.

DA gap. $\Delta H_{\mathrm{DA}} = H_{\mathrm{local}} - H_{\mathrm{verified}}$ measures the lag between the chain head and the highest height whose DA submission the validator has confirmed as retrievable from its own Celestia bridge node. This is bridge-level retrievability, not a data-availability-sampling proof; Section 6.10 describes what the prototype publishes.

P2P health. Peer count alone cannot distinguish a long-lived honest peer from a fresh Sybil, and eclipse attacks combine vectors no single metric captures [61, 62]. The profiler combines six bounds – subnet diversity, active anchor peers, clean peers, churn rate, mean peer tenure, and latency – applied to the Engram P2P interface, the Bitcoin client, and the Celestia client. These bounds characterize peripheral and partial P2P degradation; a partition of the core consensus network below a quorum is outside what any FSM can handle (Section 1).

Agreement versus correctness of health. Lemma 1 (Section 4.3) bounds what Byzantine validators can do to the aggregate; it does not make an honest sensor accurate. The six P2P bounds are a heuristic detector: an eclipse that keeps peer count, subnet diversity, tenure, and

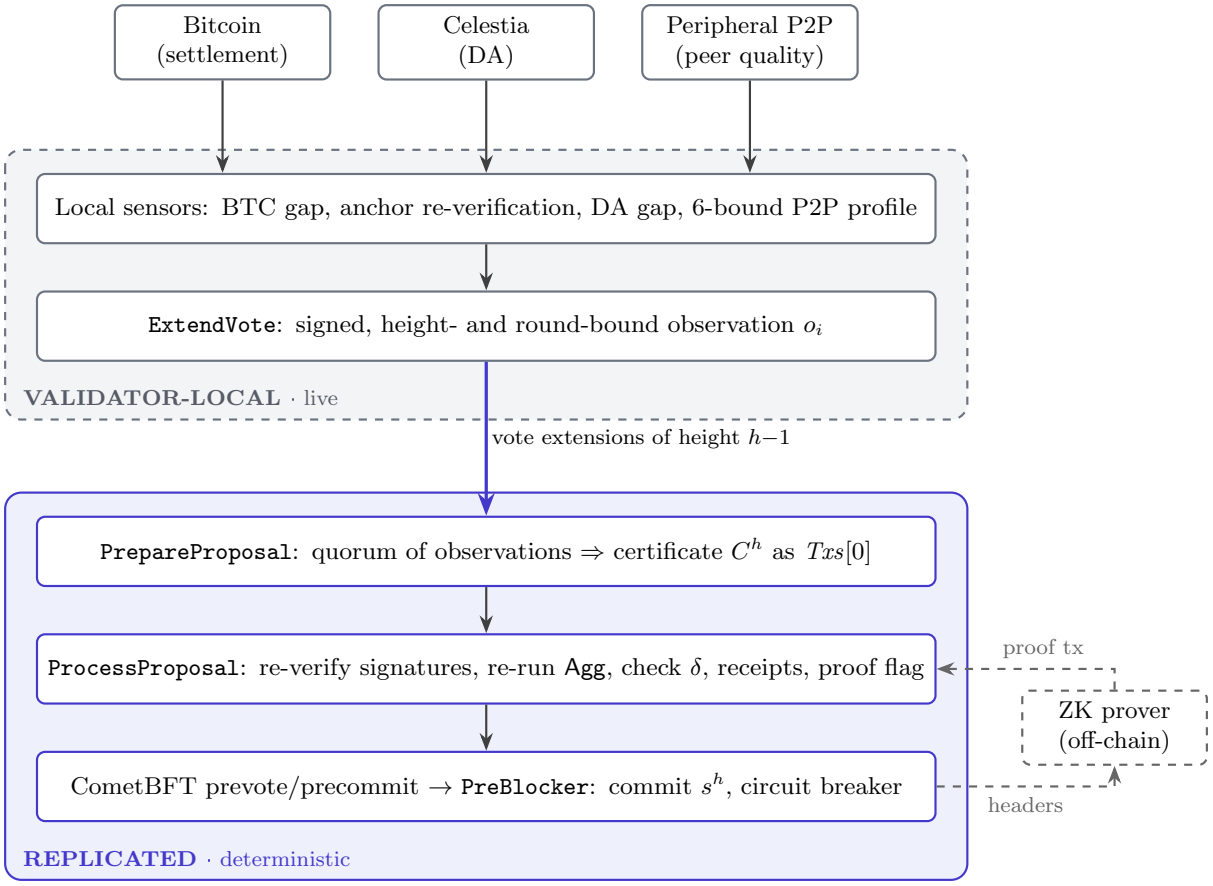


Figure 2: Engram architecture. Live sensor reads stay in the validator-local zone and enter consensus only as signed vote extensions (bold arrow, the single crossing). The replicated zone, from certificate aggregation downward, is a pure function of data every validator holds, so the committed FSM state s^h is deterministic. The dashed path is the recovery proof (Section 4.7).

latency within bounds while controlling a validator's view is not detected by them. The TLA+ model assumes that an eclipsed honest validator reports a degraded reading and never a healthier one (Section 5); that is a sensor-soundness assumption of the model, not a property derived from the six bounds.

4.3. Health Certificates

The health certificate turns asynchronous, possibly inconsistent local observations into a single value that every validator can recompute.

Observation. At height $h-1$ and round r, validator i produces an observation $o_i = (i, m_i, \sigma_i)$. Here m_i is its local metric vector (Section 4.2) – Bitcoin gap, anchored and current Bitcoin heights, anchor re-verification result, DA gap and verified height, DA sampling and attestation flags, and the six P2P-quality bounds – and σ_i is i's signature over CometBFT's vote-extension sign bytes $(chainID, h-1, r, m_i)$. The extension is produced in `ExtendVote`, the only ABCI++ hook permitted to read live sensors.

Certificate. Let $r^\star$ be the round in which height $h-1$ committed. The proposer of height h receives the extended commit of height $h-1$ and places

$$C^h = \big(h-1,\ r^\star,\ \{o_i\}_{i \in S}\big)$$

as the first transaction of its proposal. C^h is *well-formed* iff (a) every σ_i verifies against the

genesis key of validator i for $(h{-}1, r^\star)$; (b) each validator contributes at most one observation; (c) every $i \in S$ has a commit precommit in round $r^\star$ of height $h{-}1$; and (d) $\mathrm{wt}(S) > 2W/3$, so S is a quorum.

Aggregation. For a numeric metric j, the aggregate is the smallest value whose cumulative weight exceeds half of the certificate weight; a Boolean metric is true iff the weight backing true exceeds half of it:

$$\widehat{m}_j = \min\bigl\{x : \mathrm{wt}(\{i \in S : m_{i,j} \le x\}) > \tfrac{1}{2}\,\mathrm{wt}(S)\bigr\}, \qquad (2)$$
$$\widehat{b}_j = \mathbf{1}\bigl[\mathrm{wt}(\{i \in S : b_{i,j} = 1\}) > \tfrac{1}{2}\,\mathrm{wt}(S)\bigr].$$

$\mathsf{Agg}(C^h)$ collects these values. The Boolean metrics are failure flags (failed anchor re-verification, failed DA sampling, failed DA attestation), so a tie leaves the flag unset. Agg is a pure function: every validator that receives C^h computes the same value without contacting any peripheral network. The TLA+ model gives every validator the same voting power (Section 5).

Lemma 1 (honest-bounded aggregate). *Let C^h be well-formed and $\mathrm{wt}(B) < W/3$. For every numeric metric j,*

$$\min_{i \in S \cap H} m_{i,j} \;\le\; \widehat{m}_j \;\le\; \max_{i \in S \cap H} m_{i,j},$$

and every Boolean metric on which all members of $S \cap H$ agree takes that value. Proof sketch. Since $\mathrm{wt}(S) > 2W/3$ and $\mathrm{wt}(S \cap B) < W/3$, the Byzantine weight in S is below $\mathrm{wt}(S)/2$, so the honest weight in S exceeds $\mathrm{wt}(S)/2$. Let a and b be the smallest and largest honest values. The weight of values at most b is at least the honest weight, so $\widehat{m}_j \le b$. The weight of values below a is at most the non-honest weight, which is below $\mathrm{wt}(S)/2$, so no value below a satisfies Eq. (2) and $\widehat{m}_j \ge a$. For a Boolean metric on which all honest members agree on v, the weight backing v exceeds half, and the weight backing the other value is below half. □

Lemma 1 does not require honest observations to agree; it bounds what the adversary can do when they disagree. Each adversarial case reduces to it:

- *Stale or replayed observation.* A signature binds the height and round; an observation signed at another height or round fails (a) and is discarded.
- *Conflicting observations.* Two different signed observations by one validator count once by (b); Byzantine weight is below $W/3$ either way.
- *False observations.* Byzantine validators below $W/3$ reporting extreme values cannot move the aggregate outside the honest range (Lemma 1).
- *Proposer omission.* The proposer chooses S, and may omit honest observations. If S still meets (d), Lemma 1 still holds. The proposer's residual influence is choosing *which* honest value in the honest range is selected when honest validators genuinely disagree. If omission drops S to weight at most $2W/3$, the certificate is not well-formed and honest validators reject the proposal. The only exception is the first height after vote extensions are enabled, whose predecessor carries no extensions; there the certificate is replaced by a maximally degraded metric vector, the conservative choice. [PENDING: re-run E8 live with the omission attack (B3).]
- *Missing or malformed observations.* CometBFT accepts a precommit's extension only if its signature verifies under the validator's consensus key, and the application's `VerifyVoteExtension` accepts only extensions that decode; signatures are checked again against the genesis keys when the certificate is aggregated. The extended commit for $h{-}1$ carries weight above $2W/3$, so an honest proposer can always build a well-formed certificate, and a Byzantine validator cannot reduce the usable weight by sending a decodable extension with a bad signature, because CometBFT would not have accepted

that vote. This relies on CometBFT and the aggregation using the same key and sign bytes.

- *Divergent honest views.* When honest views straddle a threshold, different proposers may select certificates on different sides of it. Each resulting transition is legal and deterministic given its certificate; the FSM's hysteresis absorbs short alternations (Section 4.5).

Height and round binding. The certificate committed at height h describes observations signed at height $h-1$, one block of lag. A signature binds $(chainID, h-1, r, m_i)$, and a well-formed certificate takes observations only from the commit round $r^\star$ and only from validators whose precommit is in that commit. This prevents replay of an observation from another height or round into C^h. The binding is a property of the consensus instance, not of the measurement: a validator can still sign a cached Bitcoin tip or stale peer metrics (Section 4.2).

4.4. Finite State Machine

The FSM has four states (Table 5). ANCHORED is normal operation: blocks are valid only with a Bitcoin receipt and a DA receipt within tolerance, and withdrawals are permitted. SUSPICIOUS is an early-warning state reached after a sustained warning; withdrawals remain permitted, but admission of forced transactions is rate-limited by a bounded queue. SOVEREIGN is the fallback: the Bitcoin and DA receipt checks are relaxed for each layer that the certificate does not report healthy, blocks continue under the validator quorum, and the circuit breaker locks all cross-chain withdrawals. RECOVERING is entered once the certificate reports healthy peripherals; withdrawals stay locked until the exit gate of Section 4.7 is satisfied.

Table 5: FSM states and their validity requirements. The exit gate is defined in Section 4.7.

State	Validity requires	Withdrawals	Forced-tx admission	BTC settlement
ANCHORED	Quorum; BTC and DA receipts	Permitted	Full	Yes (K_{DEEP} conf.)
SUSPICIOUS	Quorum; receipt of each layer reported healthy	Permitted	Rate-limited	Yes, delayed
SOVEREIGN	Quorum; receipt of each layer reported healthy	Locked	Full	No
RECOVERING	Quorum; BTC and DA receipts; exit gate	Locked	Full	Pending proof

CometBFT's deterministic finality is unaffected by the FSM state; the measured commit interval on our testbed is about 1–2 s in every state. Bitcoin settlement is probabilistic and follows Bitcoin's own confirmation time (K_{DEEP} blocks, ∼10 min per block on mainnet); it is not measured in this paper.

Fig. 3 shows the transition function δ. A proposal carries a target state s^h; honest validators prevote for it only if

$$s^h = \delta\big(s^{h-1}, \mathsf{Agg}(C^h)\big), \tag{3}$$

where δ also reads the hysteresis counters committed with s^{h-1} (Section 4.5). Since Agg and δ are deterministic and the state and counters are committed, every honest validator reaches the same verdict on a given proposal, and the committed state inherits CometBFT's agreement guarantee. δ evaluates the conditions in a fixed priority: a critical condition first, then a warning (in RECOVERING, any non-healthy reading), then a healthy reading. The first condition that applies decides, so a critical condition overrides every hysteresis counter.

4.5. Transitions and Hysteresis

A *warning* (suspicious Bitcoin gap, unhealthy DA, or unhealthy P2P quality) demotes ANCHORED to SUSPICIOUS only after $k_\downarrow$ consecutive non-healthy blocks. A *critical* condition (excessive Bitcoin gap, failed anchor re-verification, complete anchor-peer isolation, or a SUSPICIOUS timeout

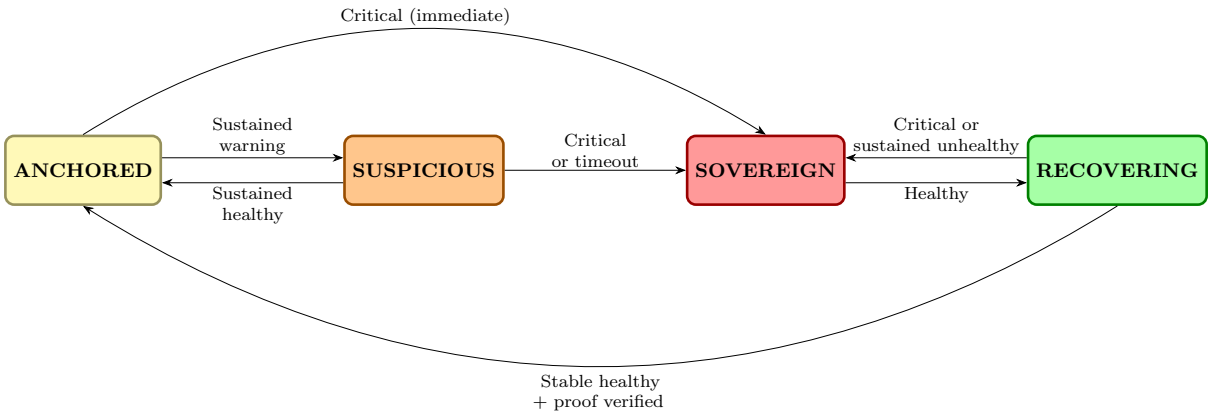


Figure 3: FSM transition function δ. *Sustained* means the condition accumulates over several blocks (hysteresis; up-counters tolerate isolated non-healthy blocks); critical conditions skip the wait and act immediately. The waits are given in Table 6.

Table 6: Hysteresis waits of the FSM (defaults of the prototype; Section 4.5).

Wait	**Applies to**	**Default**	**Adaptive**
Down wait $k_{\downarrow}$	ANCHORED→SUSPICIOUS, RECOVERING→SOVEREIGN	2, doubles per flap level up to 8	Yes
Suspicious exit $k_{\uparrow}$	SUSPICIOUS→ANCHORED	2	No
Recovery H	RECOVERING→ANCHORED	2	No

of T_{sus} blocks) moves the chain to SOVEREIGN immediately and bypasses every counter. Recovery is deliberately slower than degradation: SUSPICIOUS→ANCHORED requires $k_{\uparrow}$ consecutive healthy blocks, and RECOVERING→ANCHORED requires the safe-block counter to reach H plus the exit gate of Section 4.7. SOVEREIGN→RECOVERING needs only a healthy certificate.

To resist an adversary who repeatedly toggles a peripheral, the down threshold $k_{\downarrow}$ is adaptive. It starts at a genesis value and doubles per flap level, up to a genesis cap, and both demotion edges (ANCHORED→SUSPICIOUS and the non-critical RECOVERING→SOVEREIGN) use it. The flap level rises by one for each such demotion until the cap is reached, falls by one after T_{stable} healthy ANCHORED blocks, and is not reset by a successful recovery.

The counters follow the committed state. The unhealthy streak counts non-healthy blocks absorbed in ANCHORED or RECOVERING; it resets on any transition and on a healthy block. The safe-block counter in RECOVERING rises on healthy blocks up to H and falls by one on an absorbed non-healthy block; the corresponding SUSPICIOUS counter behaves the same way up to $k_{\uparrow}$. Any other transition resets both. Section 7.3 evaluates all three hysteresis-gated edges.

4.6. Extended Proposal and Validity Predicate

A proposal P_h at height h carries, in $Txs[0]$, the certificate C^h, the target state s^h, a health flag, a DA receipt (last DA-verified height and attestation flag), a Bitcoin receipt (checkpoint height and block hash), and, when applicable, a recovery-proof reference.

Definition 1 (Engram proposal validity). `ProcessProposal` accepts a proposal P_h iff

$$\mathsf{Valid}_h(P_h) \iff \mathsf{WellFormed}(C^h) \wedge \mathsf{StateOK} \wedge \mathsf{ReceiptsOK} \wedge \mathsf{WithdrawalsOK} \wedge \mathsf{RecoveryOK}, \quad (4)$$

where

- $\mathsf{WellFormed}(C^h)$: C^h satisfies conditions (a)–(d) of Section 4.3;

- StateOK: $s^h = \delta(s^{h-1}, \mathsf{Agg}(C^h))$ and the health flag matches $\mathsf{Agg}(C^h)$;
- ReceiptsOK: in ANCHORED and RECOVERING, and for any layer the certificate reports healthy, the DA and Bitcoin receipts are consistent with $\mathsf{Agg}(C^h)$ within the round-dependent tolerances τ_{BTC} and τ_{DA} defined below;
- WithdrawalsOK: no withdrawal is included while s^h is SOVEREIGN or RECOVERING;
- RecoveryOK: the proposal carries a recovery-proof reference with a fresh DA attestation exactly when it targets RECOVERING with the safe-block counter at H, and the exit RECOVERING→ANCHORED is taken only on a proof the chain has verified (Section 4.7).

`ProcessProposal` never reads a live sensor. The FSM state is written to replicated state only by `PreBlocker` after commit, from the certificate already agreed upon.

The tolerances are deterministic functions of the round r and genesis parameters, so every validator computes the same bounds. Each is a base threshold plus a widening w_r that is capped, so no number of round skips enlarges it beyond its maximum. A DA receipt may lag the Engram height by $\tau_{\mathrm{DA}} = \Theta_{\mathrm{DA}} + w_r^{\mathrm{DA}}$ Engram blocks, where Θ_{DA} is the DA threshold and $w_r^{\mathrm{DA}} = 0$ for $r \leq 1$, 2 for $r = 2$, and 4 for $r \geq 3$. A Bitcoin receipt's checkpoint height may lag the Bitcoin tip by $\tau_{\mathrm{BTC}} = K_{\mathrm{DEEP_FINALITY}} + 2 + w_r^{\mathrm{BTC}}$ Bitcoin blocks, where $w_r^{\mathrm{BTC}} = 0$ for $r \leq 2$ and 1 for $r \geq 3$. The base $K_{\mathrm{DEEP_FINALITY}} + 2$ is the depth a confirmed checkpoint already lags the tip plus two blocks of propagation slack. Theorem 1 takes both tolerances at the round in which B_h committed.

ReceiptsOK is exactly what a *hard-gated* modular chain enforces unconditionally. Engram's only liveness-relevant change is to make it conditional on the certified state: when the certificate reports a layer unhealthy outside ANCHORED and RECOVERING, that layer's receipt is no longer required, and the circuit breaker (WithdrawalsOK) removes the operations whose safety depended on it.

4.7. Recovery Proof and Re-anchoring

Normative protocol. The exit defined here assumes proof-only recovery: it requires a recovery proof that the chain has verified. The prototype also contains an optional engineering heuristic that can satisfy the proof condition; Section 8.3 identifies it.

Statement. On leaving SOVEREIGN, a prover builds a Noir circuit proof (Barretenberg UltraHonk backend [69,74]) over a contiguous range of $n \leq N_{\max} = 256$ headers starting from the last anchored root rt_{last}. A header is the tuple (*prev_hash*, *fsm_state*, *withdrawal_locked*, *state_root*), where *state_root* is CometBFT's application hash for that block. The circuit checks that (a) each header's *prev_hash* equals the Poseidon2 hash of its predecessor, the first links to rt_{last}, and the hash of the last header equals the claimed rt_{new}; (b) every header has *fsm_state* in {SOVEREIGN, RECOVERING} and *withdrawal_locked* set; and (c) the first and last blocks of the range carry ed25519 precommit signatures (round 0) from at least two thirds of the voting power of a validator set whose hash is a public input (the compiled circuit fixes the set size at seven). Unused slots are padding, so proving time, verification time, and proof size are constant in n (Appendix B). The circuit does not check each transition against δ, and the lock flag is the one the chain recorded in the header.

Intended verifier. A CometBFT quorum certificate shows that a quorum signed a block. The recovery proof adds a statement about the interval between two signed blocks: every recorded state in it is SOVEREIGN or RECOVERING with the lock flag set. This matters to a verifier that does not replay the chain, such as a bridge or light client on another chain, a node syncing from genesis, or a Bitcoin observer (Section 3.2). For validators, which already hold the headers, it is

a consistency check. The target verifier is an external light client that accepts a re-anchoring checkpoint only with a valid proof; the prototype does not include such a client.

Longer intervals. An interval longer than 256 headers is covered by a sequence of proofs: each accepted proof advances the on-chain rt_{last} to its rt_{new}, and the next proof must start from that root. This costs $\lceil L/256 \rceil$ proofs for L blocks. An aggregator circuit can fold up to four chunk proofs into one proof of the same size, which makes verification cost independent of L up to 1024 headers at the price of one extra proof (Appendix C).

Who verifies what. The recovery proof is submitted as a message and verified by *every Engram validator* during block execution, using an embedded, deterministic WebAssembly build of the UltraHonk verifier. The message handler binds the proof's public inputs to chain state:

- rt_{last} to the committed last anchored root, and rt_{new} to the accumulator of a header the chain recorded;
- the first height to the last anchored height plus one, and the header count to the distance between the two checkpoints;
- the first and last block hashes to the CometBFT hashes the chain recorded at those heights, and the validator-set hash to that of the genesis set.

A verified height is latched so a proof cannot be replayed. Interior headers are bound only through the accumulator, not through per-block signatures. Bitcoin only timestamps the checkpoint; it does not verify the proof. After recovery, validators resume writing `OP_RETURN` checkpoints of 44 bytes (a 4-byte tag, the Engram height, and the CometBFT block hash at that height), and the Bitcoin gap sensor requires them to reach depth $K_{\text{DEEP_FINALITY}}$. Anchor re-verification compares the hash with the local block hash. The proof's rt_{new} is a Poseidon2 accumulator, not a block hash, and is not written to Bitcoin.

Proof timing. The proof message executes in block h and latches the height of the new checkpoint. The `PreBlocker` of block $h{+}1$ recomputes the proof-validity flag from that height, and `ProcessProposal` reads this committed flag when it evaluates the exit edge, so RECOVERING→ANCHORED cannot occur before block $h{+}2$. `ProcessProposal` does not verify the proof itself.

Resulting guarantee. The recovery proof guarantees, to anyone who verifies it, that (i) the recorded headers of the SOVEREIGN interval form a single hash chain between two boundary blocks signed by at least two thirds of the voting power, and (ii) every recorded state is SOVEREIGN or RECOVERING with the lock flag set. Because the lock flag is the chain's own record, (ii) does not prove that no withdrawal executed. The proof does not make the interval's *ordering* secure against a Byzantine supermajority: during SOVEREIGN, safety rests on the validator quorum alone, which is precisely why withdrawals are locked. Re-anchoring restores Bitcoin timestamping from the recovery point forward.

Data availability of the Sovereign interval. A recovery proof shows the interval is recorded consistently, not that its data is *available*. The exit condition is therefore stronger than the proof: RECOVERING→ANCHORED requires (1) a recovery proof that the chain has verified and (2) a DA backfill certificate – a statement, backed by the quorum of signed observations in the health certificate, that the block data for every height of the SOVEREIGN interval has been published to the DA layer and is retrievable. During SOVEREIGN, each validator keeps the interval's blocks in its own store (they are ordinary committed CometBFT blocks) and queues them for publication; on recovery the queue is drained in height order. Each validator reports a contiguous height: the highest height such that every height up to it has been confirmed retrievable. The certificate takes the weighted median $\widehat{d}$ of these reports. Let e be the height at which RECOVERING was entered, and let safe^{h-1} and proof^{h-1} be the committed safe-block counter and proof-validity

flag. The exit edge is taken at height h iff

$$\text{healthy}\big(\mathsf{Agg}(C^h)\big) \ \wedge\ \text{safe}^{h-1} = H \ \wedge\ \text{proof}^{h-1} \ \wedge\ \widehat{d} \geq e - 1. \tag{5}$$

The DA gap sensor keeps measuring head liveness, so the DA layer stays healthy while the backfill drains. The specification abstracts the backfill condition as a Boolean, `da_backfill_valid`. The prototype implements the queue and the gate; Section 6.10 describes the remaining live validation.

5. Security and Formal Analysis

This section states the properties that follow from the mechanisms of Section 4 under the assumptions of Section 3, and reports what bounded model checking establishes about them.

5.1. Analytical Properties

Security of mode switching. A transition happens only when a well-formed health certificate implies it (Lemma 1, Section 4.3). Byzantine validators below $W/3$ cannot move a numeric aggregate outside the range of the honest observations in a well-formed certificate, and cannot overturn a Boolean value on which all represented honest validators agree. When honest observations straddle an FSM threshold, the proposer's choice of quorum subset can still influence which side of the threshold the aggregate falls on; hysteresis limits this influence but does not remove it. The second attack surface is griefing by an adversary that degrades the peripheral view of honest validators (Section 8).

Theorem 1 (withdrawal safety). *Assume* $\text{wt}(B) < W/3$*, no Bitcoin reorganization deeper than* $K_{DEEP_FINALITY}$*, and an honest DA layer. Assume further that (M1) honest validators reject a block that contains a withdrawal while the state is* Sovereign *or* Recovering*; (M2) each Bitcoin checkpoint commits the chain through its height, and with it the replicated state; and (M3)* Recovering→Anchored *requires a recovery proof that the chain has verified and a DA backfill certificate for the whole* Sovereign *interval. If a withdrawal commits in block* B_h*, then* P_{agree} *holds, the state at* h *is* Anchored *or* Suspicious*, and there are Engram heights* c_s *and* $c_a \geq h - \tau_{DA}$ *such that* $P_{settle}(c_s)$ *and* $P_{avail}(c_a)$ *hold, where the checkpoint that settles* c_s *lags the certified Bitcoin tip by at most* τ_{BTC} *Bitcoin blocks. Every height up to* $\min(c_s, c_a)$ *is settled and available, including every* Sovereign *interval that precedes* B_h*.*

Proof. P_{agree} follows from $\text{wt}(B) < W/3$. By (M1), the state at h is Anchored or Suspicious. In the transition function δ, Suspicious is entered only from Anchored, and Sovereign is left only through Recovering; so every Sovereign interval before h ended with a Recovering→Anchored transition, which by (M3) required a verified recovery proof and a DA backfill certificate. Hence the interval is valid, hash-linked, and available. In the Anchored and Suspicious blocks since then, every block carries receipts within τ_{BTC} and τ_{DA}. The Bitcoin receipt in B_h attests a checkpoint at depth at least $K_{\text{DEEP_FINALITY}}$ for some c_s; by (M2) it commits the chain through c_s, so rewriting the prefix up to c_s requires a deeper reorganization, which is excluded. The DA receipt in B_h attests data availability up to some $c_a \geq h - \tau_{\text{DA}}$, and the DA layer is honest. □

Theorem 1 guarantees settlement only through c_s and availability only through c_a. The blocks after them may not yet have either guarantee when the withdrawal commits, and a withdrawal may depend on state created inside that exposure window; the theorem does not tie a withdrawal to a dependency height. Liveness comes from relaxing the receipt checks in Sovereign; user safety comes from tying withdrawals to the settled and available prefix.

The prototype realizes the three assumptions as follows:

- (M1) is enforced in the validity predicate and the ante handler.
- (M2) holds because each checkpoint payload carries the CometBFT block hash at its height c. That hash commits the header of B_c, hence the chain of blocks through c by the hash links, and the header carries the application hash of the state before B_c. The checkpoint therefore commits that state and the transactions of B_c, and, since execution is deterministic, the state after B_c.
- (M3) is enforced on the exit edge by the backfill condition, which the FSM safety run checks as an action property, and by the proof condition (qualified in Section 8.3).

The theorem is a pen-and-paper argument over these assumptions, not a mechanized statement about the implementation.

5.2. Formal Model

Model-checking consensus, FSM, and sensors together is intractable at useful scale. We therefore specify the protocol in TLA+ [70] as four layers – an abstract consensus object, a Tendermint-style consensus layer, the FSM layer, and a server layer composing them – and check each layer with TLC and, for bounded safety searches, Apalache [42]. The target property is: under partial synchrony and an f-bounded adversary, no two honest validators commit conflicting blocks or conflicting FSM states, and any such conflict implies double-signing evidence.

Safety versus liveness. The linking invariant *QuorumOverlap* – any two valid quorums share at least one honest validator – is a *safety* premise: together with honest validators' refusal to sign conflicting votes, it prevents two conflicting decisions. It does not establish liveness. Following LiDO [47], liveness is argued separately, and in our specification it rests on explicit fairness assumptions that the model checker takes as hypotheses:

- *Partial synchrony:* GST is reached infinitely often (`[]<>GSTReached`), and after GST messages between honest validators are delivered (weak fairness on each node's `Push` and `Pull`).
- *Timeout progress:* weak fairness on round-timeout expiry and on the $f+1$ round-skip (fast-forward) action.
- *Proposer progress:* weak fairness on each honest node's invocation step, so every honest proposer whose round is reached after GST eventually proposes.
- *FSM progress:* weak fairness on the FSM transition and on environment updates, so a persistently healthy (or unhealthy) certificate eventually drives the corresponding transition.

Under these hypotheses TLC checks the FSM liveness properties directly. A TLC ablation that removes the $f+1$ fast-forward, which would show the timeout-progress assumption is necessary, has not completed: its best run reached depth 12 (69,408,583 states generated, 828,673 distinct) with no violation. The necessity of that assumption is therefore not established here.

5.3. Verification Scope

What is checked. The evidence has two levels.

- *Per-layer bounded model checking (exhaustive within each stated bounded configuration).* Each layer's own safety invariants and liveness properties are checked in isolation at its own bounded configuration (Table 7), together with the ablation study (Table 8).
- *Cross-layer refinement (open).* The obligation that the composed server layer refines the abstract consensus object – which is what would transfer the target property to the

composed system – is specified at three scales (C1: $N{=}4, f{=}1$; C2: deeper rounds; C3: $N{=}7, f{=}2$) but has not converged at any of them. The composed system is therefore *not* formally verified end-to-end; its whole-system evidence in this paper is empirical (Section 7).

To make C1 tractable we split it by which actions may fire and check each part with TLC under symmetry reduction over the three honest validators. With the consensus core frozen, the environment part (sensor updates, clock ticks, DA backfill) completed: 17,500 distinct states, depth 10, 10 m 35 s, no violation. With the environment frozen, the consensus-core part did not complete: it was stopped after 1 h 41 m at 19,848,141 distinct states (depth 15, 13,542,792 states still queued) with no violation. Neither part covers the interleaving of environment and consensus steps, and no composition argument is established, so C1 is not claimed.

The server layer carries two guards that stop it from certifying a timeout for a round an honest validator has already left, and one added abstract action for a round ended by a quorum of nil precommits. These are changes to the checked model, and the last weakens the target specification. An inductive check of every server action with Apalache would avoid the composition gap; it has not been attempted.

Health certificates in the model. The server layer models the certificate. Each validator's observation holds three Booleans (healthy, warning, critical) and the Bitcoin and DA gaps clipped to 0..3, written to `sensor_view` by `ServerExtendVote`. A Byzantine validator may sign any observation; an honest validator with an eclipsed P2P layer may report a degraded one, never a healthier one, which is a sensor-soundness assumption of the model. Every validator has the same voting power. The proposer may choose any member set that forms a quorum. The aggregate (`AggregateHealthCertificate`) is a per-field majority for the Booleans and a weighted median for the gaps, and `IsValidProposal` rejects a certificate below a quorum.

Lemma 1 is checked as `AggWithinHonestRange` with Apalache, with the observation vector left symbolic so that one check covers every vector. For $N{=}4$ and $f{=}1$ it reported no violation (22 s). With two Byzantine validators in the same network it reports a violation (25 s), so the bound on Byzantine members is necessary. Signatures, height and round binding, and the individual P2P counters are not modelled. The check covers the aggregate as a function; the certificate's role inside `IsValidProposal` is covered only by the open refinement obligation.

5.4. Safety Invariants and Attack Resilience Lemmas

Six state invariants are specified and model-checked.

- *Circuit Breaker Isolation:* withdrawals are locked if and only if the state is SOVEREIGN or RECOVERING.
- *Hysteresis Integrity:* RECOVERING→ANCHORED requires the full hysteresis wait and a valid recovery proof.
- *Suspicious-Exit Hysteresis Integrity:* the corresponding guard on SUSPICIOUS→ANCHORED.
- *Strict FSM Transition Safety:* only transitions drawn in Fig. 3 occur.
- *Monotonicity Safety:* heights and time never decrease.
- *Sovereign-Data Publication:* every RECOVERING→ANCHORED transition occurs with the DA backfill condition true. This is an action property, checked in the FSM safety run; the Server-layer drivers have not been re-run with it.

Five attack-specific properties are enforced by the validity predicate: resistance to data withholding; resistance to forged or reorganized anchors through anchor re-verification; a deterministic

bound on Byzantine timeout flooding; resistance to eclipse-driven recovery through the six-bound P2P gate; and accountability, i.e. any safety violation implies double-signing evidence. Accountability is inherited from CometBFT's equivocation evidence pool rather than proved anew. All of these properties are conditional on the Bitcoin reorg-depth bound of Section 3.1.

5.5. Bounded Results and Mechanism Ablation

Table 7: Per-Layer, Non-Refinement Verification Results (TLC)

Layer / property	States gen.	Distinct	Depth	Time	Workers	Viol.
Tendermint (safety)	3,772,809	537,536	25	1 m 01 s	10	0
FSM (safety)	1,371,377,696	4,542,271	23	10 m 19 s	10	0
FSM (liveness)	17,986,105	31,105	23	3 m 45 s	10	0

Each row is one completed TLC run of that layer's driver. Depth is the number of breadth-first levels at which the search terminated with every state satisfying the state constraint explored, so it is the diameter of the constrained state graph and not a cap. Hardware: Apple M1 Pro, 10 cores, 32 GB RAM.

Table 7 reports the per-layer results, each obtained by running that layer's own driver standalone at the configuration of Table 9. All three runs found zero violations in their bounded state spaces. They cover the Tendermint-style and FSM layers; the abstract and server layers are exercised only through the open refinement obligation. The ablation study spans two layers: the hysteresis and P2P health gate belong to the FSM layer, the circuit breaker and DA consistency to the Tendermint-layer driver, so each result holds within its own layer's configuration.

Table 8: Ablation Study: Removing One Mechanism at a Time (Apalache)

Removed	Targeted threat	Depth	Time	Violated property
Circuit breaker	Withdrawal leakage	2	66.7 s	`Sanity_NeverAttempt WithdrawalLeakage`
Hysteresis	Premature re-anchoring	6	117.0 s	`Sanity_NoIllegal HysteresisExit`
P2P health gate	False recovery under eclipse	3	24.2 s	`Sanity_NoIllegal P2PRecovery`
DA consistency	Data withholding	2	63.3 s	`Sanity_Never ProposeWithheldData`

Table 8 reports the ablation study. Each row removes one mechanism and re-runs verification against the sanity property that mechanism protects. Apalache's bounded SMT search found each counterexample in 24–117 s at depth 2–6. Each ablation reproduced the violation its mechanism is designed to prevent: without the circuit breaker a withdrawal leaks; without hysteresis the chain re-anchors prematurely; without the P2P gate an eclipsed validator forces an illegal recovery; and without DA consistency a withholding proposal is accepted. Single-factor ablation does not test joint removal, so it does not establish that the set of mechanisms is sufficient.

6. Implementation and Evaluation Methodology

6.1. Implementation

The reference implementation is a Cosmos SDK application, `engramd`, on a forked CometBFT core. The application mounts no `x/auth`, `x/bank`, or `x/staking`: transactions are unsigned envelopes and the validator set comes from the genesis file. Three modules hold the protocol logic: `x/sovereignty` (the FSM, the sensors, the circuit breaker, and the ABCI++ hooks), `x/da` (the Celestia receipt and its verification), and `x/anchor` (the Bitcoin settlement receipt

Table 9: TLC Run Configurations

Run	Constants and bounds
Tendermint (safety)	N=4, f=1 (three honest, one Byzantine node); MaxRound 2; MaxBTCHeight 10; MaxEngramHeight 10; MaxTimestamp 2; symmetry reduction over the three honest nodes
FSM (safety, liveness)	SuspiciousThreshold 2; SovereignThreshold 4; DAThreshold 2; RecoveryHysteresisWait 2; SuspiciousHysteresisWait 2; MaxSuspiciousTime 2; DownHysteresisThreshold 2; MaxDownHysteresisThreshold 8; MinPeers 3; MinSubnetDiversity 2; MinAnchorPeers 1; MaxChurnRate 10; MinAvgTenure 100; MaxPeerLatency 50; the Bitcoin and Engram heights stay below 10 (safety run); no symmetry reduction

and its re-verification). The ABCI++ hooks follow the pipeline of Section 4.1: only `ExtendVote` reads live sensors, and `PreBlocker` commits the FSM transition from the agreed certificate.

The fork differs from upstream CometBFT in four ways: an $f+1$ timeout round-skip, per-peer round-trip-time measurement and a peer health monitor that feed the P2P sensor, per-peer rate limiting and bounded round acceptance for timeout messages, and the proposal round passed to the application through `PrepareProposal` and `ProcessProposal`.

Bitcoin and Celestia are live dependencies, deployed as described in Section 6.2. The recovery proof is a Noir circuit proved with Barretenberg (Section 4.7); a prover process watches for a SOVEREIGN backlog, builds the proof from committed headers, and submits it as a transaction that every validator verifies during block execution. Section 6.3 lists the versions.

6.2. Testbed

The testbed runs the reference implementation of Section 6.1 as seven validators with equal voting power (n=7, f=2), each in its own container. Validator gossip uses a dedicated full mesh of pairwise links, separate from the network that carries Bitcoin, Celestia, and RPC traffic; this separation lets us degrade peripheral connectivity without partitioning the core consensus network. The peripheral layers are real software, not simulators: a two-node Bitcoin regtest network mined continuously at roughly one block per 20 s, and a Celestia network with seven bridge nodes, one per validator, so that each validator forms its own DA view. The eclipse and Sybil rows (A1 and A2) of Appendix A predate the move to seven validators and ran on a four-validator cluster (f=1). [PENDING: re-run A1/A2 at n=7, and run the f=2 collusion suite of Section 7.1.]

6.3. Environments and Software Versions

All results come from one environment, Testbed A: the seven-validator Docker deployment above on a single host. Every table and figure in Section 7 is a Testbed A measurement. Table 10 lists the configuration.

6.4. Baselines

Hard-gated modular baseline (primary, RQ2). The same binary with the FSM pinned to ANCHORED: condition (3) of the validity predicate (Section 4.6) is enforced at every height, so a proposal is valid only if it carries a Bitcoin receipt and a DA receipt within tolerance. This is the behavior of a chain that makes settlement and DA inclusion part of block validity, and it isolates the effect of Engram's adaptive relaxation. It is a runtime mode of the same binary (`ENGRAM_MODE=hard-gated`); Section 7.2 runs the RQ2 fault schedule against it.

Table 10: Testbed Configuration and Software Versions

Component	Testbed A
Host	Apple M1 Pro (10 cores: 8P + 2E), 32 GB RAM, 1 TB SSD; macOS 27.0
Container runtime	Docker 29.0.1; VM allocation 10 CPUs, 23.4 GiB RAM
Validators (7)	No CPU or memory limit per container
Consensus core	CometBFT v0.39.3 module path, replaced by the fork `engram-consensus-core` at commit `493a20121`
Application	Cosmos SDK v0.54.3; this repository at [PENDING: results commit]
Bitcoin	Bitcoin Core v25.0, regtest, two nodes (0.3 CPU, 512 MB each)
Celestia	celestia-app v1.11.0 (0.5 CPU, 1 GB); seven celestia-node v0.14.1 bridges (0.3 CPU, 512 MB each)
Prover	nargo 1.0.0-beta.22, `bb` 5.0.0-nightly.20260522 (6 CPU, 8 GB limit)
Fault injection	Pumba (image tag `latest`) driving kernel `tc netem`
Formal tools	TLC 2026.04.18 (rev. c45c125); Apalache v0.58.3 on JDK 17

CPU and memory figures are container limits. The fork's differences from upstream CometBFT are listed in Section 6.1.

Soft baseline (RQ2). The real FSM runs, but withdrawals are never locked in SOVEREIGN or RECOVERING (`ENGRAM_MODE=soft`). It keeps liveness and drops the circuit breaker, which isolates what the withdrawal lock contributes.

Vanilla CometBFT (control, RQ4). The same application with the health-aware ABCI++ hooks disabled. It does not depend on Bitcoin or Celestia and therefore never stalls when they fail; it serves as the cost reference for RQ4 and as a control showing that measured stalls or fallbacks are not artifacts of the harness.

Ablation baselines. A peer-count-only P2P detector isolates the contribution of the six-bound profiler (RQ1), and each hysteresis edge with its threshold set to its floor isolates the contribution of hysteresis (RQ3).

6.5. Fault Injection

Network-level faults use kernel traffic control (`tc netem` via Pumba): uniform delay and jitter, packet loss, single-node isolation, and a heterogeneous per-validator wide-area profile approximating seven regions. Appendix E lists the parameters of every profile. The *WAN latency baseline* is the per-validator delay and jitter profile of Table 27, held for the whole run unless a step in Table 11 suspends it.

Application-level faults reproduce mechanisms that delay alone cannot, and each simulates the consequence of a failure rather than its cause.

- *Bitcoin congestion* suspends the submission of new checkpoints (a pending one still confirms), so the finality gap grows directly. It does not congest the Bitcoin mempool.
- *DA outage* stops one or all seven Celestia bridge containers. It does not emulate a data-availability-sampling failure.
- *Byzantine behavior* is injected through attacker processes or through a validator configured to forge proposal fields, omit transactions, double-sign, or flood timeout messages.

Every fault targets a peripheral layer or a minority of validators; none partitions the core consensus network below a quorum, consistent with the scope defined in Section 1.

6.6. Fault Schedules

Table 11 gives the onset, mechanism, and duration of each fault in the two scripted runs of Section 7.2. Times are relative to the start of each run, durations are the length of the polling

window the script holds the fault for, and each of S2–S5 is followed by a wait of up to 90 s for the validators to return to ANCHORED before the next scenario starts.

Table 11: Fault Schedules of the Scripted Runs

Step	Fault and mechanism	Duration
Composite run (standing WAN latency baseline)		
C1	Baseline, no fault	15 s
C2	Bitcoin fault: 500 ± 100 ms RPC delay on one `bitcoind` (profile length 2 min)	60 s
C3	All seven Celestia bridges stopped, added to C2	60 s
C4	Three cycles of 5% packet loss on validators 1 and 2 (WAN baseline suspended)	$3 \times$ (20 s on, 10 s off)
C5	Hold; bridges still stopped	30 s
C6	Heal: Bitcoin profile cleaned up, bridges restarted	30 s
C7	Wait for ANCHORED (limit 600 s)	until recovery
Seven-scenario sweep		
S1	Healthy baseline	15 s
S2	Checkpoint submission suspended on all validators	180 s
S3	All seven Celestia bridges stopped	60 s
S4	300 ± 50 ms delay on the six validator links of validator 1	120 s
S5	100% packet loss on validator 1	180 s
S6	One `bitcoind` and all seven bridges stopped	90 s
S7	Faults removed; wait for ANCHORED (limit 600 s)	until recovery

6.7. FSM Parameters

FSM parameters are fixed in the genesis file, so every validator holds identical values. All live experiments use the defaults of Table 12 unless a section states an override. Sensors are sampled once per block, when a validator signs its vote extension; the measurement scripts poll each validator's RPC every 3 s (2 s during the packet-loss cycles of C4). Blocks arrive about once per second.

6.8. Synthetic Workloads

Noise profile (in-process). The hysteresis sweeps drive the real state machine with sensor readings generated per block. Each block is disturbed independently with probability 0.2 and healthy otherwise, for 100 blocks, from a fixed pseudo-random seed (7 for the RECOVERING→ANCHORED sweep, 11 for the ANCHORED→SUSPICIOUS sweep), so every threshold value sees the same noise sequence. A disturbance sets sensors to a failing reading for one block. Critical disturbances set the Bitcoin gap to `SovereignThreshold`. Warning-level disturbances set the Bitcoin gap to `SuspiciousThreshold`, mark DA unavailable, or give the P2P sensor a snapshot with no clean peers, diversity, or tenure and only the minimum anchor count. The combined environments apply several at once.

Monte Carlo P2P study. A *snapshot* is one synthetic tuple of the six P2P sensor fields (subnet diversity, active anchors, clean peers, churn rate, mean tenure, latency). A benign snapshot draws each field inside its healthy range with a uniform random margin. An attack of *intensity* $x \in [0, 1]$ perturbs the fields it targets in proportion to x (for example, the relay attack adds x (`MaxPeerLatency`+150) ms of latency and leaves the other fields benign); $x = 0$ is benign and $x = 1$ is the full attack. Detectors are the six-bound profiler and the peer-count-only rule CleanPeers $\geq$ `MinPeers`. The study uses production-scale P2P thresholds: 8 subnets, 4 anchors, 20 clean peers, churn 5, tenure 300 s, latency 200 ms. Four measurements per cell follow, all from fixed seeds.

- *False-positive rate:* the share of 2,000 benign snapshots flagged unhealthy.

Table 12: FSM Parameters Used in the Live Experiments

Parameter	Value	Unit and role
`SuspiciousThreshold`	5	Bitcoin gap, blocks
`SovereignThreshold`	8	Bitcoin gap, blocks
`DAThreshold`	30	DA gap, blocks
`KDeepFinality`	2	Reorg depth, Bitcoin blocks
`MinPeers`	3	Clean peers
`MinSubnetDiversity`	2	Distinct subnets
`MinAnchorPeers`	2	Active anchor peers
`MaxChurnRate`	5	Peer churn events
`MinAvgTenure`	300	Mean peer tenure, s
`MaxPeerLatency`	200	Round-trip time, ms
`RecoveryHysteresisWait`	2	RECOVERING→ANCHORED, blocks
`DownHysteresisThreshold`	2	ANCHORED→SUSPICIOUS, blocks
`MaxDownHysteresisThreshold`	8	Cap on the backed-off threshold
`MaxUnprovenTailBlocks`	128	Tail after the proved checkpoint that still counts (default 8)
`ProofOnlyRecovery`	true	Recovery requires a verified proof (default false)
`StableWindow`	16	Healthy ANCHORED blocks that lower the flap level by one
`SuspiciousHysteresisWait`	2	SUSPICIOUS→ANCHORED, blocks
`MaxSuspiciousTime`	24	SUSPICIOUS escalation bound, blocks

Overrides: the Docker testbed runs the table's `DownHysteresisThreshold` of 2 with the backoff cap of 8, so the down threshold is adaptive, and sets `SuspiciousHysteresisWait` to 8, which is not adaptive. The hysteresis sweeps of Section 7.3 vary the down threshold, a static value being `MaxDownHysteresisThreshold` equal to `DownHysteresisThreshold`; the Monte Carlo study of Section 7.1 uses the P2P thresholds of Section 6.8.

- *False-negative rate:* the share of 2,000 full-intensity snapshots flagged healthy.
- *Detection delay:* over 500 ramps, the number of snapshots until the first flag, with intensity rising $0.1, 0.2, \ldots, 1.0$ over 10 snapshots. The mean is taken over ramps that were detected.
- *Evasion threshold:* the largest intensity (on the 0–1 scale) at which a majority of 20 probes still read healthy, found by a 12-step binary search that queries the detector, so it bounds an adversary with oracle access.

The ramp-shape study uses 300 ramps per shape and the three shapes x (linear), $\sqrt{x}$ (front-loaded), and x^2 (slow). The upper 95% confidence bound on a rate observed as 0 in n trials is about $3/n$ (rule of three): 0.15% for $n = 2{,}000$ and 1% for $n = 300$. [PENDING: report the standard deviation of detection delay per cell.]

6.9. Metrics

Let a run record the FSM state s_b of every block b in a window W.

- *Anchored uptime* $= |\{b \in W : s_b = \text{ANCHORED}\}|/|W|$. In-process, W is the whole recorded timeline: for the noisy environments, the 100 noisy blocks plus the setup blocks that precede them (two for the RECOVERING→ANCHORED sweep, one for the others). The no-noise control of that sweep ends at the first ANCHORED block, so its window is only the setup blocks plus the wait, and its uptime falls as the wait grows. Live, W is the sequence of RPC samples of one validator over a 300 s window.
- *Flapping count* is the number of transitions that immediately reverse the preceding transition, for example ANCHORED→SUSPICIOUS followed by SUSPICIOUS→ANCHORED, over the same W. Live runs detect reversals between consecutive samples, so a reversal faster than the polling interval is not seen.

- *Absorption rate* is $a/(a+d)$, where a is the number of disturbed blocks that begin in ANCHORED and leave the state unchanged, and d is the number that move it to SUSPICIOUS. It is defined only on the ANCHORED→SUSPICIOUS and SUSPICIOUS→ANCHORED sweeps.
- *Blocks/s* $= (h_{\max} - h_{\min})/T$ over a measurement window of length T (60 s after an 8 s warm-up in Section 7.4). *Steady blocks/s* $= 1/\bar{\imath}$, where $\bar{\imath}$ is the mean block interval. *Gap fraction* $= 1 - n\bar{\imath}/T$, where n is the number of observed intervals: the share of the window not accounted for by intervals at the steady pace, that is, time spent in heights that took longer than one interval.

6.10. Sim-to-Real Gap

Four properties of the testbed limit how its results transfer to a production deployment.

- *Single physical host.* All Testbed A containers share one machine's CPU and memory, so resource contention can affect every validator at once; wide-area conditions are emulated with `tc netem` rather than real geographic distribution.
- *Accelerated Bitcoin.* Regtest blocks arrive about 30× faster than on mainnet. Height-based thresholds are unaffected, but wall-clock durations scale accordingly.
- *DA payload.* The DA publisher submits the transactions of committed blocks in height-ordered batches of at most 1.5 MB and tracks the highest contiguous published height; the exit gate of Section 4.7 requires that height to cover the SOVEREIGN interval. The gate rests on a quorum-reported contiguous height, not on a cryptographic proof of retrievability. Section 7.3 reports the republication of the interval after a live DA outage.
- *Single-bridge trust.* Each validator trusts its own bridge node, which downloads and verifies full Celestia blocks, rather than sampling across independent peers; our DA-outage results measure bridge failures, not sampling-detected partial withholding.

7. Evaluation

Each research question is answered by the live experiments that bear on it most directly; the remaining attack scenarios, the prover-backend comparison, and the Bitcoin-reorg boundary test appear in Appendices A–D. Unless stated otherwise, all numbers are live measurements on Testbed A, the seven-validator testbed of Section 6 (Table 10); FSM parameters are those of Table 12, and metrics follow Section 6.9.

7.1. RQ1: Safety Under Byzantine Behavior

Single Byzantine proposer. The most direct attack on the extended consensus object is a proposer that claims an FSM state not implied by the certificate. One of the seven validators (n=7, 14.3% of the voting power) was configured to do so. Every honest validator recomputed $\delta(s^{h-1}, \mathsf{Agg}(C^h))$, rejected the forged claim in `ProcessProposal`, and the chain committed only honest proposals, with zero AppHash divergence. The same outcome held for the nine further single-actor attacks of Appendix A (eclipse, Sybil, data withholding, forged Bitcoin receipt, withdrawal in SOVEREIGN, censorship, a combined attack, double-signing, and timeout flooding).

Colluding Byzantine validators (f=2). The threat model allows f=2 colluding validators at n=7. We therefore define a collusion suite in which two validators act jointly: (B1) both report coordinated false health observations; (B2) one reports false observations while the other, as proposer, builds its certificate to favor them; (B3) the proposer omits honest observations, both down to the smallest quorum (five of seven validators) and below it; (B4) both double-sign at the same height; (B5) both attempt to include a withdrawal while in SOVEREIGN; and (B6) the

proposer attaches a stale or forged Bitcoin or DA receipt. AppHash agreement is necessary but not sufficient evidence of safety, since honest validators could agree on an illegal transition. For every committed height we therefore also replay $\delta(s^{h-1}, \mathsf{Agg}(C^h))$ offline from the committed certificate and check that it equals the committed s^h, that the aggregate lies within the honest range of Lemma 1, and that no withdrawal commits while $s^h \in \{\textsc{Sovereign}, \textsc{Recovering}\}$. [PENDING: run B1–B6 at n=7 and report per-case AppHash agreement, semantic-replay violations, and the honest-range check.]

Peripheral P2P quorum gate. A P2P detector is useful for safety only if its signal cannot be abused by a minority. We first compare the six-bound profiler (Section 4.2) with a peer-count-only detector in a Monte Carlo study (2,000 benign and 2,000 attack snapshots plus 500 ramps per cell; Section 6.8), then run the two most diagnostic attacks live.

Table 13: P2P Detector Accuracy: Six-Bound Profiler vs. Peer-Count-Only (Synthetic; 2,000 benign and 2,000 attack snapshots, 500 ramps per cell)

	Six-bound profiler		**Peer-count-only**	
Attack	**Delay**	**Evasion**	**Delay**	**Evasion**
P1 Peer-slot exhaustion	2.23	0.17	never	1.00
P2 Multi-subnet Sybil	1.00	0.00	never	1.00
P3 Churn-based rotation	1.70	0.12	never	1.00
P4 Relay-node latency	2.71	0.25	never	1.00

Delay: mean snapshots to first detection over a 10-snapshot ramp ("never": no ramp was detected). Evasion: largest attack intensity on the 0–1 scale at which the detector still reads healthy. Both detectors have 0/2,000 false positives (95% upper bound 0.15%). The profiler has 0/2,000 false negatives at full intensity; the peer-count-only detector has 2,000/2,000, since each attack preserves the clean-peer count.

Table 13 shows that the profiler detects every attack at every intensity an adaptive adversary could choose (evasion threshold ≤ 0.25), under three ramp shapes (linear, front-loaded, slow; 300/300 ramps detected per cell, 95% upper bound on the miss rate 1%) and regardless of concurrent Bitcoin or DA degradation. The peer-count-only detector detects none of the 300 ramps in any cell. The live runs show what happens after detection. In P3 (churn-based rotation: eight restart cycles on two of seven validators, 29% of voting power), the churn metric on both attacked validators crossed its threshold, yet no FSM transition was committed: two validators' signed observations cannot move a certified aggregate backed by a quorum. In P4 (a sustained 350±50 ms delay on all of one validator's P2P interfaces, observed by all seven validators), the chain moved $\textsc{Anchored}\to\textsc{Suspicious}\to\textsc{Sovereign}$ at +39 s and +72 s, held $\textsc{Sovereign}$ for the 244-second attack, and returned to $\textsc{Anchored}$ within 5 s of the delay clearing (Fig. 4).

P3 and P4 are two outcomes of one mechanism. The same kind of sensor-confirmed degradation is ignored when a minority observes it and drives a bounded cascade when all validators do, as Lemma 1 predicts: a minority cannot move the certified aggregate, whereas a degradation seen by every honest validator must move it.

7.2. RQ2: Liveness Under Peripheral Degradation

Engram versus a hard-gated modular chain. The hard-gated baseline (Section 6.4) enforces Bitcoin and DA receipts at every height; the soft baseline keeps Engram's FSM but never locks withdrawals. We ran each system through three faults: Bitcoin checkpoint submission suspended, all seven Celestia bridges stopped, and both. A trial is 60 s healthy, 180 s of fault and 120 s of recovery, with a withdrawal-marked transaction sent every 2 s; each cell of Table 14 has three trials. The hard-gated chain is redeployed for every trial because it does not recover on its own (below). Recovery uses proof-only re-anchoring with an unproven-tail bound of 128 blocks (Section 4.7).

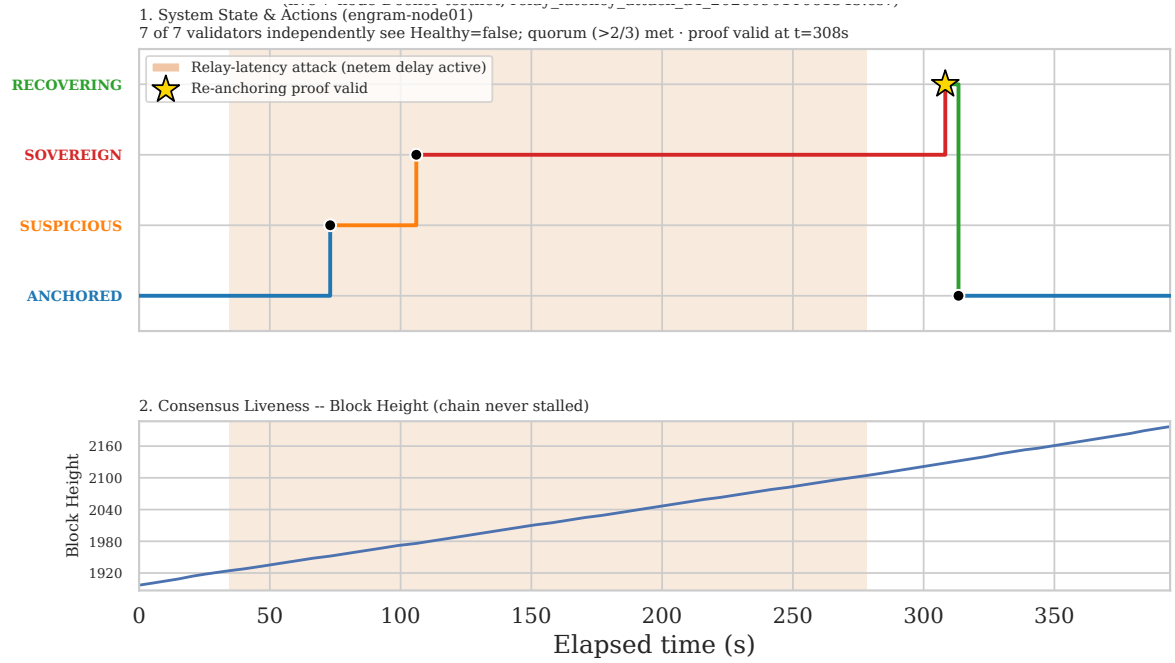


Figure 4: FSM cascade under a real, quorum-visible P2P relay-latency attack (P4): all 7 validators independently sense the degradation, the transition commits once quorum is met, and block production (bottom panel) never stalls through the cascade or the recovery.

Under a DA outage the hard-gated chain commits 23 blocks in 180 s, 16% of its healthy rate, with proposals reaching round 21, and commits nothing in the 120 s after the layer returns in all three trials; the combined fault is the same. Under checkpoint suspension it slows to 56% (74 blocks, round 12): the receipt tolerance widens with the round up to a cap, so stale checkpoints are admitted for a while, but it then stops resuming as well. In the one trial observed beyond the recovery window, the chain was still stuck after twelve minutes, with proposals rejected at round 57 and above. The stall persists because the checkpoint a proposal needs is derived from blocks the chain can no longer commit, so in this implementation it takes a manual restart. Engram commits 147–153 blocks (96–102% of its healthy rate) under every fault, with a longest gap of at most 2.5 s and rounds 0–1, and resumes within two seconds of the fault ending.

The withdrawal columns separate the systems on safety. After the circuit breaker has locked, Engram committed none; this count is a consistency check of the breaker, because the window opens when Engram's own FSM reaches Sovereign. The hard-gated chain, which has no breaker, committed 3 withdrawals after that moment under DA faults and 6–9 under checkpoint suspension, few only because it commits few blocks. Engram's cost shows in the last column: it committed 18–19 withdrawals before locking under DA faults and 45–56 under checkpoint suspension, because it needs about 41 s and 117 s to leave Anchored for Sovereign. The soft baseline shows the other side of the trade-off. It runs Engram's FSM and keeps Engram's liveness, 98–99% of the healthy block rate with a longest gap of at most 2.5 s, but it never locks, so it committed 31 withdrawals after the lock moment under checkpoint suspension and 71–72 under DA faults, against none for Engram. Counted by the FSM state at each height, 87–97 of its 175 withdrawals under checkpoint suspension and 132–137 of about 180 under DA faults were committed while the state was Sovereign or Recovering. Only Engram combines the liveness of the soft chain with the zero count of a chain that locks.

Composite fault. Engram was then run through a continuous, scripted composite-fault sequence under a standing wide-area latency profile (Table 11): a Bitcoin fault, then a full DA

Table 14: Liveness and withdrawals under peripheral degradation (180 s fault, n trials per cell, mean and range). Rate is the fault-window block rate as a share of the same trial's healthy rate; "No resume" counts trials in which the chain committed no block in the 120 s after the fault ended ("—" in the gap column: it never resumed). The last two columns count withdrawals committed after and before the moment Engram's circuit breaker locks, 117 s (checkpoint pause), 41 s (DA outage) and 43 s (both) after the fault begins; the same window is applied to every system.

			Liveness in the fault					Withdrawals committed	
Fault	**System**	n	Blocks	Rate	Longest gap (s)	Max round	No resume	After lock	Before lock
Checkpoint pause	Hard-gated	3	74 (73–74)	56 (53–57)%	—	12	3/3	8 (6–9)	55 (54–56)
	Engram	3	147	102%	2.4 (2.4–2.5)	1 (0–1)	0/3	0	52 (45–56)
	Soft	3	144 (143–145)	99 (98–100)%	2.5 (2.4–2.5)	1 (0–1)	0/3	31 (30–31)	56 (56–57)
DA outage	Hard-gated	3	23 (22–24)	16 (14–17)%	—	21 (21–22)	3/3	3	19 (18–20)
	Engram	3	150 (147–153)	96 (93–99)%	1.4 (1.3–1.4)	0	0/3	0	18 (18–19)
	Soft	3	151 (149–152)	99 (98–100)%	1.5 (1.4–1.7)	0	0/3	72 (72–73)	19 (19–20)
Both	Hard-gated	3	24 (23–24)	17 (16–19)%	—	21 (21–22)	3/3	3	19 (19–20)
	Engram	3	153 (150–155)	99 (97–104)%	1.4 (1.3–1.4)	0	0/3	0	19 (18–20)
	Soft	3	151 (150–152)	98 (97–99)%	1.3	0	0/3	71 (70–71)	20 (20–21)

outage (all seven bridges stopped), then three cycles of packet loss on the validator P2P network, healed in turn. The packet loss degrades peer quality without removing quorum connectivity. Fig. 5 shows the resulting timeline.

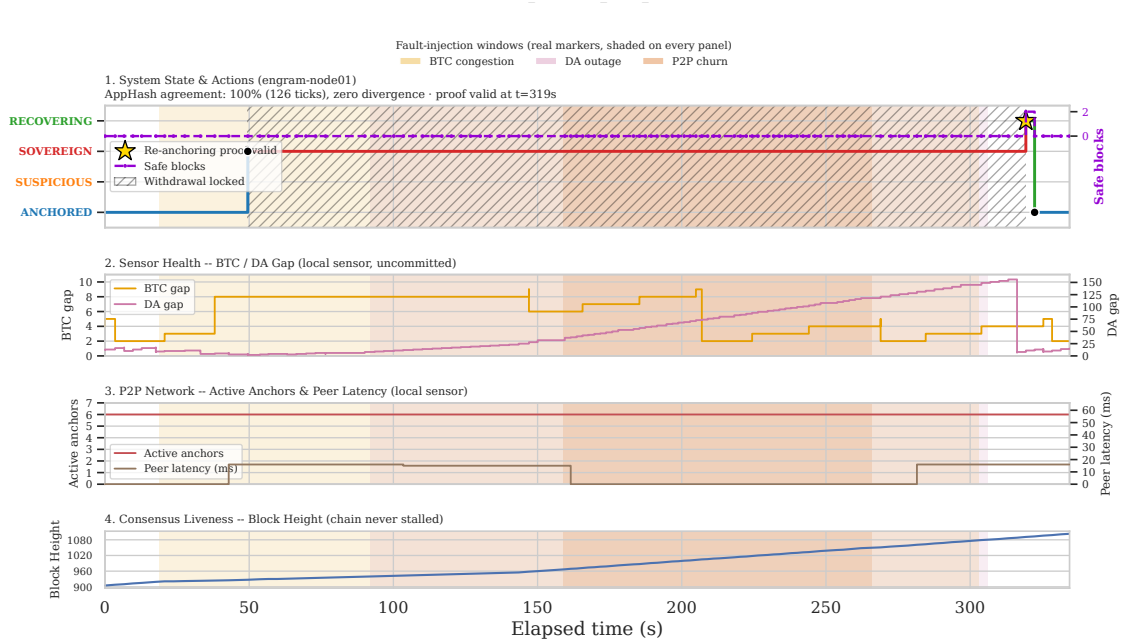


Figure 5: FSM state timeline, sensor readings, P2P network state, and block height under a layered composite fault sequence (Bitcoin congestion + DA outage + P2P churn, overlapping at their peak).

Over the 341-second run (742 samples, 126 polling ticks, one trial), the height advanced continuously from 905 to 1093 with no stalled height, and all seven validators agreed on the AppHash at every tick (zero divergent heights, zero RPC errors). The chain entered SOVEREIGN under the combined fault and, after the faults were healed, the recovery proof was verified at t=319 s, the chain entered RECOVERING at t=321 s and ANCHORED at t=324 s. Withdrawals stayed locked throughout SOVEREIGN and RECOVERING, as the circuit-breaker invariant requires. These timings come from a single trial. [PENDING: repeat the composite and seven-scenario

runs (≥ 5 trials) and report recovery time as mean $\pm$ 95% CI.]

Seven-scenario sweep. The composite trace does not isolate which failure mode drives which behavior. A second run therefore walks the chain through seven sequential scenarios in one 864-second run (one trial; Table 11): S1 healthy baseline, S2 Bitcoin congestion (checkpoint submission suspended), S3 DA outage (bridges stopped), S4 partial P2P eclipse, S5 anchor-peer isolation of one validator, S6 combined Bitcoin and DA failure, and S7 recovery. Each scenario starts from the state the previous one left; 153 FSM transitions were logged across the seven validators (Fig. 6).

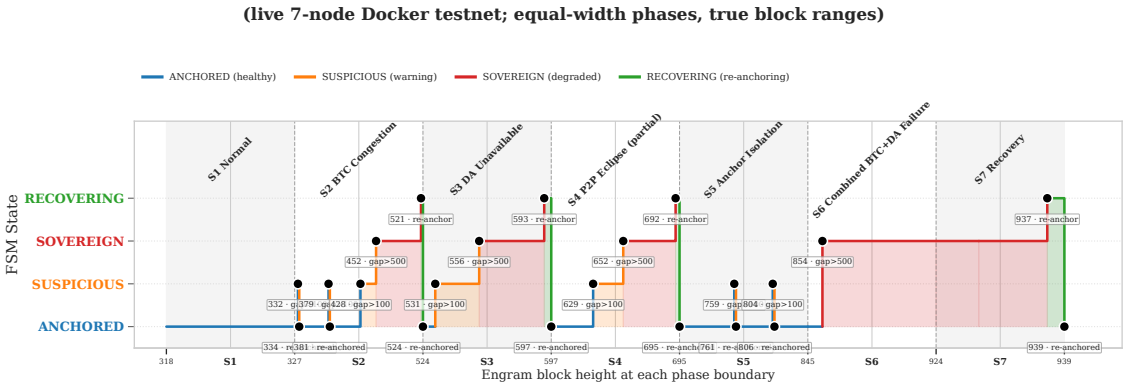


Figure 6: Continuous FSM-state timeline across seven sequential fault-injection scenarios (S1–S7), true block-height ranges, live 7-validator testbed. Each scenario begins where the previous scenario's recovery ended.

Each scenario produced the expected response. S2 and S3 each escalated through SUSPICIOUS to SOVEREIGN and recovered. S4 escalated to SOVEREIGN, whereas S5 produced only a transient SUSPICIOUS excursion absorbed by hysteresis: isolating one of seven validators from its anchor peers leaves the certified active-anchor-peer count above zero, so the critical condition never holds. S4 and S5 are therefore distinct failure classes rather than two intensities of one signal. S6 escalated directly to SOVEREIGN, and S7 returned to ANCHORED through a submitted and verified recovery proof. Two further observations stand out. First, S2 contains two brief SUSPICIOUS excursions (19–22 s and 77–80 s) that return to ANCHORED before the escalation at 141 s, showing hysteresis absorbing live rather than synthetic noise. Second, block production stayed close to baseline: block-interval means ranged 1.30–1.93 s across all seven scenarios (vs. 1.34 s at the S1 healthy baseline; all intervals in this paragraph are Engram's), and the largest p95 intervals occurred in S4 (4.27 s) and S7 (3.10 s). Engram keeps committing blocks under overlapping faults and under each failure mode in isolation, and returns to ANCHORED in every case.

7.3. RQ3: Recovery and Flapping

Hysteresis should suppress oscillation ("flapping") without unduly delaying legitimate transitions. We swept the threshold on each of the three gated edges – RECOVERING→ANCHORED, ANCHORED→SUSPICIOUS, and SUSPICIOUS→ANCHORED – in-process under fixed-seed synthetic noise and live on the seven-validator cluster under real network noise (Fig. 7).

On the ANCHORED→SUSPICIOUS edge (Table 15), raising the down-hysteresis threshold from 1 to 4 collapses flapping from 31 events to 1 and raises anchored uptime from 61.4% to 98.0%, and the absorption rate saturates near 100% by threshold 4–6. The live spot-check changes the conclusion. Under wide-area jitter alone, with no injected fault, thresholds 1, 2, and 4 still flapped, and only thresholds ≥ 6 were clean (0 flaps, 100% uptime). The default of 2 used in the

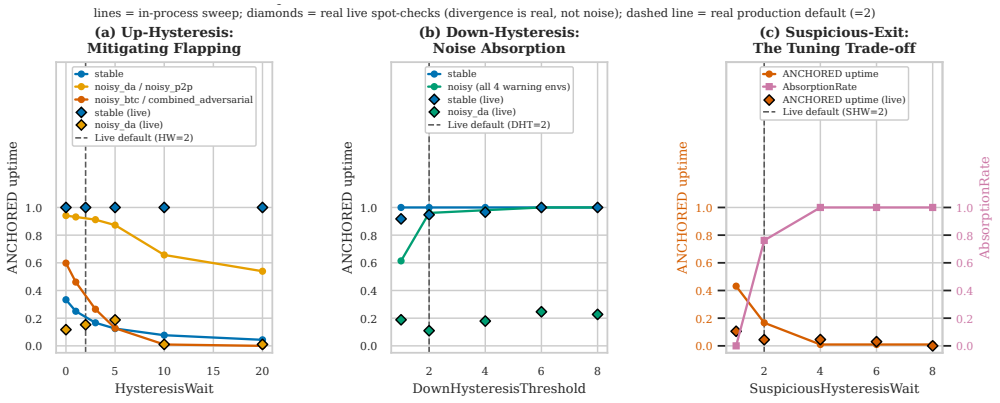


Figure 7: Hysteresis parameter sweep across all three absorb edges. Lines: in-process sweep under different noise profiles; diamonds: live spot-checks on the real 7-validator cluster (divergence from the in-process line is real network behavior, not measurement noise); dashed line: the protocol's live production default.

Table 15: Down-Hysteresis Sweep (ANCHORED→SUSPICIOUS), In-Process (101 blocks, noise probability 0.2, seed 11, one run per threshold)

Threshold	1	2	4	6	8
Flapping (count)	31	1	1	0	0
Anchored uptime (%)	61.39	96.04	98.02	100.00	100.00
Absorption rate (%)	0.00	94.74	95.24	100.00	100.00

other experiments is therefore too low for WAN conditions; the other experiments did not count flaps and cannot be taken as evidence to the contrary. A static default must trade reaction time (low k) against noise immunity (high k), and the right point depends on network conditions that change over time.

The other two edges show the opposite side of the trade-off. On the RECOVERING→ANCHORED edge, a larger hysteresis wait strictly reduces anchored uptime under sustained critical-level noise (59.8%→0.0% as the threshold goes from 0 to 20), under the critical-level noise of Section 6.8. This follows from the design. In RECOVERING, a critical reading returns the chain directly to SOVEREIGN (the RECOVERING branch of `CalculateNextState`), so the run of healthy blocks must start again, and a longer wait is less likely to complete between two critical readings. On the SUSPICIOUS→ANCHORED edge, absorption rate rises from 0% to 100% as the threshold goes from 1 to 4, at a cost: the SUSPICIOUS timeout keeps running while the counter accumulates, so only thresholds 1–2 reached ANCHORED before the timeout forced SOVEREIGN. Any adaptive threshold must therefore stay below the SUSPICIOUS timeout, which bounds $k_{\max}$.

Adaptive hysteresis. We therefore extend the backoff already used on RECOVERING to the down threshold $k_{\downarrow}$ of both demotion edges (Section 4.5). The effective threshold is $k_{\downarrow} = \min(k_{\max}, k_0 \cdot 2^c)$, where the flap level c rises by one at each such demotion and falls by one after T_{stable} healthy ANCHORED blocks. $k_{\downarrow}$ depends only on committed state, so it remains deterministic across validators. It starts at the responsive value k_0 and rises only when the network proves noisy, which also raises the cost of the repeated-trigger attack discussed in Section 8. The mechanism is implemented in the specification and in the prototype. We swept the down threshold live on the seven-validator cluster (Table 16): static $k \in \{2, 4, 6, 8\}$, each its own cap, and the adaptive threshold (k_0=2, $k_{\max}$=8), under wide-area latency jitter with no fault, under DA blips (all seven bridges stopped and restored in cycles of tens of seconds), and under a sustained DA outage (all bridges stopped) to measure detection. All 33 runs completed.

A static value keeps no hidden state, so one genesis served all its trials and environments; the adaptive configuration was redeployed for every run because its flap level lives on the chain. Static k=2, k=6 and the adaptive threshold have three trials each; k=4 and k=8 have one.

Table 16: Live down-edge sweep on seven validators (300 s runs, mean ± 95% Student-t interval over n trials, none for n=1; flaps and uptime averaged over the seven validators of a trial). "Uptime" is the share of the run in ANCHORED; "Detection" is the delay from stopping the bridges of the sustained outage to the first non-ANCHORED state on one validator, and "To SOVEREIGN" the delay to SOVEREIGN.

Configuration	n	Flaps (WAN)	Uptime (WAN, %)	Uptime (DA blips, %)	Detection (s)	To Sovereign (s)
static k=2	3	9.0 ± 0.0	82 ± 3	1 ± 0	9.4 ± 0.4	38 ± 2
static k=4	1	0.0	100	1	11.1	41
static k=6	3	0.0 ± 0.0	100 ± 0	2 ± 2	12.8 ± 3.2	44 ± 5
static k=8	1	0.0	100	3	14.6	43
adaptive (2 → 8, T_{stable}=16)	3	8.3 ± 2.9	85 ± 7	1 ± 0	11.7 ± 0.9	42 ± 3
adaptive (2 → 8, T_{stable}=64)	3	3.0 ± 0.0	93 ± 2	–	11.1 ± 1.9	41 ± 3

Under jitter alone the static default of 2 flapped nine times per validator and spent 82% of the run in ANCHORED, and every static value from 4 up held ANCHORED without a flap. The adaptive threshold did not behave like the larger values: 8.3 flaps and 85% uptime, within the interval of k=2. Under DA blips no configuration stayed in ANCHORED (1–3% uptime), so the threshold did not matter there. Detection delay grew with the threshold, from 9.4 s at k=2 to 14.6 s at k=8 (n=1), and the adaptive configuration sat between k=2 and k=6 at 11.7 s. In this sweep the adaptive threshold therefore bought no flap suppression over a static k=2, whereas a static k of 4 or more suppressed all flapping at a cost of under 2 s of detection delay for k=4; with one trial at k=4 and k=8 this is an observation, not an estimate. We did not measure the cost the adaptive threshold imposes on a repeated-trigger attacker, which is its stated purpose, and we do not know why the flap level did not rise enough under jitter; the protocol's stable window of 16 blocks is one candidate we have not tested.

Appendix D complements these results by measuring how deep a real Bitcoin reorganization must be before the FSM reacts.

What ends Recovering. The state exposes whether the proof condition holds, not which path set it, so we ran one 120 s DA outage per setting and read the prover's log. With the heuristic fallback enabled, the chain left RECOVERING after 36 s while the prover's single proof was rejected as invalid (the interval had grown while it was built): the fallback, not the proof, ended recovery. With proof-only recovery and the default bound of 8 unproven tail blocks, the chain never left RECOVERING, not even after genesis (observed for over thirty minutes in two deployments), although the chain accepted the prover's proofs (seven in one deployment). A 256-header proof takes about two minutes to build and submit, more than 90 blocks at 1.2 s, so the tail was always longer than the bound. With the bound raised to 128 blocks the chain left RECOVERING 356 s after entering it, the first proof being accepted 353 s in (n=1). The bound must therefore exceed the proving latency measured in blocks, and the blocks of that tail are protected by consensus alone. No block of these intervals was committed in a round other than 0. The RQ2 trials above use proof-only recovery with the bound of 128; the composite run, the seven-scenario run and the data-availability runs used the fallback.

Data availability of the Sovereign interval. The exit condition of Section 4.7 requires the interval's block data to be republished before RECOVERING→ANCHORED. We stopped all seven Celestia bridges for 120, 300 and 600 s, restarted them, and after the chain returned to ANCHORED re-fetched every height of the interval from Celestia through `blob.GetAll`, comparing the bytes with the Engram block. Table 17 shows that all 2,291 heights over nine runs were found and identical. The chain entered SOVEREIGN 46–49 s after the bridges stopped, and the time in RECOVERING did not grow with the interval: 37–61 s for 75–77 blocks and 49–61 s for 465–475

blocks. The check is external to the chain: the gate itself rests on the quorum-reported contiguous height, and these runs show that height was truthful when all validators were honest. They cover a total DA outage on one host with n=3 per length. Runs under the earlier 512 MiB bridge limit are excluded; at that limit the bridges crash-looped and the 600 s outage never returned to RECOVERING. These runs kept the heuristic fallback of Section 4.7 enabled (proof-only recovery off), so the time in RECOVERING does not isolate the cost of the recovery proof; the data-availability results are independent of that choice. A gate that always opens would pass the runs above, so we also withheld data. A test hook makes a validator publish only head markers: its DA liveness stays healthy, so the chain can reach RECOVERING, but its contiguous height never advances. The certified height is a weighted median over the certificates, so the gate should hold when four or more of the seven validators withhold data and open when three or fewer do. Table 18 shows that it did in all seven runs. In a third control, two validators signed a contiguous height of 10^{12} while the five honest ones withheld data; the median stayed at zero and the gate held, as it should against f=2 misreporting validators. These controls use n=2–3 runs per setting, each on a freshly deployed cluster, and the withholding validators are a test configuration, not a modelled adversary. A first attempt at the third control ran with only two of its five withholding validators because the override file omitted three nodes; it is excluded.

Table 17: Republication of the SOVEREIGN interval after a total DA outage (n=3 per outage length; range over runs)

Outage (s)	Interval (blocks)	Time in Recovering (s)	Heights found, identical
120	75–77	36.6–61.1	229 / 229
300	217–219	60.9–67.3	655 / 655
600	465–475	49.0–61.2	1407 / 1407

Table 18: Backfill-gate controls after a 120 s DA outage (validators withholding block data; the gate must hold for 300 s or open)

Setting	Runs	Interval (blocks)	Outcome
4 of 7 withhold data	3	201–208	held in RECOVERING 3 / 3
3 of 7 withhold data	2	67–72	left RECOVERING after 34 and 43 s; data identical
2 inflated claims + 5 withhold	2	202–208	held in RECOVERING 2 / 2

7.4. RQ4: Cost

Where the cost is not. The extended object adds a near-constant +228 bytes per proposal (live: 247–270 bytes across states, 100% of 376 sampled blocks), and the one proof-carrying block observed was 14.9 KB, consistent with the 14.7 KB proof. Validating an accepted proposal, including the ed25519 verification of seven certificates, costs 253 μs per block (mean of ten benchmark runs, standard deviation 4 μs; 150 μs with four validators); verifying a recovery proof, a one-off event, takes 26 ms and is not included. `ExtendVote`, which reads the live Bitcoin and Celestia sensors before a validator can precommit, took 16.6 ms on average (p95 51 ms, n=5 repeats). None of these is large enough to explain a lower block rate at a ∼1 s block interval.

Where the cost is. Against a single local process, Engram's blocks arrive 0.23 s later (1.262 against 1.034 s). Table 19 shows that this gap is not Engram's. Seven vanilla validators on the same hosts, network, genesis and load produce a block every 1.233 s with vote extensions disabled and every 1.229 s with empty extensions, within 0.03 s of Engram (95% CIs of 0.015–0.023 s). Hence the 0.2 s is the price of running a seven-validator BFT network on one host, and vote extensions as such add no measurable time. Engram's own addition is about 0.03 s per height (about 2%), consistent with the 16.6 ms `ExtendVote` time. In all four arms every height

Table 19: Overhead of the Extended Consensus Object vs. Vanilla CometBFT (20 tx/s, 60 s per repeat, mean ± 95% CI)

	Vanilla			Engram
Metric	1 process	7 val., no ext.	7 val., ext. on	7 val.
Repeats n	5	5	3	5
Rounds per height	1.000	1.000	1.000	1.000
Proposals rejected / height	0	0	0	0
Nil precommits / height	0	0	0	0
Block interval, mean (s)	1.034 ± 0.001	1.233 ± 0.015	1.229 ± 0.023	1.262 ± 0.020
Block interval p50 / p95 (s)	1.032 / 1.042	1.220 / 1.313	1.217 / 1.300	1.241 / 1.335
Blocks/s	0.967	0.811	0.814	0.793
Committed tx/s	20.00	20.00	20.00	20.00
Tx commit latency p50 / p95 (s)	0.67 / 1.16	0.99 / 1.57	0.99 / 1.56	1.03 / 1.63
Mempool backlog, mean / max	8.2 / 24	14.5 / 31	14.6 / 31	15.5 / 44
Tx per block (mean)	17.6	21.1	21.1	21.5

Per-height values come from scanning *every* committed height (commit round, nil precommits, block time). Nil prevotes are not stored on chain; rejected proposals (distinct height/round/reason in the validators' logs) are the proxy for why a validator would prevote nil. "ext. on" keeps vote extensions enabled in the genesis with empty extensions; "no ext." is a fresh genesis without them. The Engram arm is a fresh deployment; a first deployment gave 1.242 ± 0.005 s (n=3). Transaction counts exclude the health certificate in `Txs[0]`. Latency runs from submission to the first 0.25 s poll that sees the transaction in a block. The 1-process arm is a local single node.

committed in the first round with no nil precommit and no rejected proposal, so extra rounds do not contribute at 20 tx/s without faults. A mean over n=5 cannot resolve a difference of a few milliseconds. The difference to the seven-validator arm without extensions is 0.029 s; combining the two reported 95% intervals (0.020 and 0.015 s) gives an approximate 95% interval of 0.004–0.054 s (0.3–4.4%). The data show an addition of a few percent at most, not a more precise value. The arms ran on separate deployments except the first Engram and the extensions-on arm, which shared one chain (Engram first); the Engram arm was repeated on a fresh deployment to check for an order effect and moved by 0.02 s.

Load. Table 19 is measured at 20 tx/s. Both systems accepted every submitted transaction into the mempool across an earlier 5–100 tx/s ramp, but mempool acceptance is not throughput. To look for the committed-throughput ceiling we offered 20, 50, 100, 200 and 500 tx/s for 60 s with five repeats per rate and drained the mempool after each (Table 20). Vanilla CometBFT and Engram committed every offered transaction at every rate, so neither saturated and the ceiling was not reached on this host; the limit we hit first was CPU, with the busiest validator at 64–83% at 500 tx/s on a host that runs all seven. The p95 latency separates the systems: vanilla stays at 1.5–1.9 s, while Engram rises from 2.2 s at 100 tx/s to 6.5 s at 500 tx/s.

The Engram figures above 100 tx/s are not a steady ANCHORED state. The peer-latency sensor read round-trip times of up to 1.2 s between validators under this load in the first Engram arm, above the 200 ms bound, so we repeated the sweep with the bound raised to 5000 ms. The chain still left ANCHORED: the share was 0.83 at 100 tx/s, 0.13 at 200 tx/s and 0 at 500 tx/s, so the peer-latency bound is not the only reason, and we did not identify the others. That arm had lower latency than the default one (p95 1.9 against 3.8 s at 200 tx/s) and was still above vanilla. We therefore cannot separate the cost of the consensus extension from the effect of a demoted FSM above 100 tx/s on this host, and report only that Engram's latency grows with load where vanilla's does not. All seven validators share the host's cores, so absolute values do not transfer to separate machines. The 5000 ms bound was set through the environment of that deployment, whose genesis we did not keep; the first Engram arm did not record the FSM state.

Table 20: Open-loop throughput, seven validators on one host (60 s per repeat, five repeats; mean ± 95% Student-t interval). The ANCHORED share is the fraction of two-second samples of one validator's FSM state spent in ANCHORED and was recorded for the third arm only.

Offered (tx/s)	Arm	Committed (tx/s)	p95 latency (s)	Peak CPU (%)	Anchored share
20	Engram	20 ± 0	1.6 ± 0.1	10 ± 2	–
20	Vanilla	20 ± 0	1.6 ± 0.1	8 ± 3	–
50	Engram	50 ± 0	1.7 ± 0.0	15 ± 8	–
50	Vanilla	50 ± 0	1.6 ± 0.0	12 ± 2	–
100	Engram	100 ± 0	2.2 ± 0.4	29 ± 6	–
100	Engram, peer-latency bound 5000 ms	100 ± 0	1.7 ± 0.1	21 ± 5	0.83 ± 0.09
100	Vanilla	100 ± 0	1.6 ± 0.1	19 ± 11	–
200	Engram	200 ± 0	3.8 ± 0.8	81 ± 11	–
200	Engram, peer-latency bound 5000 ms	200 ± 0	1.9 ± 0.5	39 ± 23	0.13 ± 0.37
200	Vanilla	200 ± 0	1.5 ± 0.1	23 ± 5	–
500	Engram	500 ± 0	6.5 ± 0.8	83 ± 12	–
500	Engram, peer-latency bound 5000 ms	500 ± 0	4.6 ± 1.1	73 ± 11	0.00 ± 0.00
500	Vanilla	500 ± 0	1.9 ± 0.4	64 ± 30	–

8. Discussion and Limitations

8.1. Security Envelope and Trade-offs

The fallback trades cross-chain withdrawals for in-chain liveness. We discuss the trade along three axes: liveness inside the chain, safety of cross-chain assets, and the cost of griefing.

Liveness inside the chain. In SOVEREIGN, block production, in-chain transactions, application logic, and the forced-transaction queue continue. Cross-chain withdrawals stop. A hard-gated chain under the same degradation commits only a fraction of its blocks and does not resume after the layer returns (Section 7.2).

Safety of cross-chain assets. The prototype has no bridge and no real assets: the application mounts neither `x/auth` nor `x/bank`, transactions are unsigned envelopes, and a withdrawal is recognized by the substring `TX_WITHDRAWAL` in the transaction bytes. The circuit breaker is therefore evaluated as a policy: it shows that the decision to admit or reject withdrawals is agreed by all validators and applied identically, not that any economic loss is prevented.

Cost of griefing. An adversary that degrades the peripheral view of *all* honest validators – for example by delaying their Bitcoin connectivity – can drive the chain into SOVEREIGN and freeze withdrawals for as long as the degradation lasts (P4, Section 7.1). This is a new attack surface relative to a chain that ignores its periphery, but not relative to a hard-gated chain: the same attacker halts the hard-gated chain entirely, whereas Engram freezes only withdrawals. Lemma 1 prevents Byzantine validators from causing the freeze alone: honest validators whose peripheral view is degraded must supply the rest of the majority. Together with the Byzantine validators they must hold more than half of the certificate weight, and the proposer's choice of quorum subset sets that weight. At n=7 with two colluders, an honest proposer that includes all seven observations requires two degraded honest validators, and a Byzantine proposer that selects a minimal quorum of five requires one. In a design without a quorum-backed health input, one node's view suffices. The freeze is the intended response to an unverifiable periphery, not a safety violation. Its economic cost to users and the attacker's cost of sustaining it are not quantified.

Repeated triggering of Suspicious. Backoff on RECOVERING already raises the price of forcing repeated failed recoveries. Without the adaptive threshold of Section 7.3, the ANCHORED↔SUSPICIOUS boundary has no such protection beyond the bounded forced-transaction queue; the adaptive threshold is meant to close this asymmetry, but the live sweep of Section 7.3

did not measure an attacker's cost, and under jitter alone it suppressed flapping no better than a static threshold of 2.

Proposer bias within the honest range. When honest observations straddle a threshold, a proposer can choose which honest value the certificate selects (Section 4.3). This cannot produce a transition that no honest view supports, but a sequence of Byzantine proposers can bias the timing of transitions. A persistent split straddling an FSM threshold (e.g. three versus four validators on each side) has not been tested live; the RQ4 scan shows no extra rounds in the unfaulted case, and the split case itself remains untested.

Restart after a long chain. The publisher's contiguous height is kept in memory. A validator that restarts therefore republishes the chain from height 0 and reports DA as failed until it has caught up. In one run, four validators restarted on a chain of about 7,400 heights; the certified DA health stayed failed and the chain remained in SOVEREIGN for more than half an hour. Persisting the contiguous height would remove the dependence on chain length; the prototype does not.

History divergence at re-anchoring. During SOVEREIGN, the ordering of blocks rests on the validator quorum alone. A coalition exceeding f could produce a SOVEREIGN history that differs from one it would have produced under Bitcoin timestamping; the recovery proof certifies the history's internal legality and quorum signatures, not its uniqueness beyond the f bound. The withdrawal lock limits what such a history can extract.

Data availability of the Sovereign interval. During SOVEREIGN, the interval's block data is held by the validators' own block stores, so its recoverability depends on those validators until it is published. The backfill gate of Section 4.7 makes publication a precondition for returning to ANCHORED.

8.2. Interpretation of Results

The formal and empirical results cover complementary ground. The TLA+ results establish, per layer and within bounded configurations, that the FSM's safety invariants hold and that removing any of four defensive mechanisms reproduces the violation it targets; Lemma 1 bounds the certified health input by honest observations. The live evaluation shows that the composed implementation behaves as specified under real peripheral failures. Because the cross-layer refinement is open, the whole-system claim rests on the empirical side. Single-trial live results (composite fault, seven-scenario sweep, P4, reorg depth) show that correct behavior occurs without characterizing its distribution; the repeated experiments (hysteresis sweeps, the n=5 overhead benchmark, the 2,000-trial detector study) do characterize distributions.

Unexpected results. Three results need a causal account.

- *Block rate.* Against a single local vanilla process, Engram's block interval is longer, which at first suggested a cost of the extended object. Seven vanilla validators on the same hosts give the same interval, with vote extensions on or off, so the gap is the cost of the seven-validator network and Engram's own addition is about 2% (approximate 95% interval 0.3–4.4%; Section 7.4).
- *Hysteresis threshold.* The default threshold of 2 flapped under wide-area jitter, and only thresholds ≥ 6 were clean (Section 7.3). We attribute this to latency variance in the Bitcoin sensor crossing the threshold in consecutive samples; a run with the jitter profile removed would test this.
- *Reorg depth.* Reorgs of depth 3–5 orphaned confirmed blocks without a reaction, while depth 15 escalated through the suspicious-duration bound (Appendix D). Whether the

anchor-reverification flag was ever set is not established, because the per-validator sensor value was not captured; logging it and sweeping depths 6–14 would locate the boundary.

8.3. Limitations

- *Testbed.* All results come from one host with accelerated Bitcoin and single-bridge DA trust; the backfill gate rests on a quorum-reported contiguous height, not a proof of retrievability (Section 6.10).
- *Proof gate.* The recovery-validity flag holds if a verified proof was submitted within the permitted unproven tail of the tracked tip, or if a heuristic holds: the last submitted Bitcoin checkpoint has reached the anchored depth. A protocol parameter disables the heuristic, so that only a verified proof satisfies the gate. The RQ2 trials used proof-only recovery with an unproven-tail bound of 128 blocks; the composite run, the seven-scenario run and the data-availability runs of Section 7.3 were made with the heuristic enabled, so their time in RECOVERING does not isolate the cost of the recovery proof.
- *Byzantine scale.* The live attacks reported so far involve one Byzantine actor at a time, and A6 and A1/A2 ran at n=4; the f=2 collusion suite (Section 7.1) addresses this.
- *Fixed validator set.* The prototype assumes one validator set throughout a degraded interval (Section 3.1).
- *Formal scope.* The cross-layer refinement is open, and the health certificate is modelled with Boolean observations in the server layer; numeric aggregation is covered by Lemma 1 only (Section 5).
- *Load range.* Load was tested up to 100 tx/s; behavior at higher rates is not characterized.

8.4. Threats to Validity

Internal validity. Host sharing can confound cost measurements. Repeated measurements with confidence intervals, and a vanilla comparison on the same host under the same load, mitigate this; the host still affects absolute values.

External validity. Results at seven validators on one host need not extrapolate to larger or geographically distributed validator sets; they are evidence of correct behavior and of the source of the overhead, not production performance figures.

Construct validity. AppHash agreement and height progression are the conventional operational proxies for safety and liveness in CometBFT-family systems. AppHash agreement alone cannot detect an illegal transition agreed on by all validators; the semantic replay check of Section 7.1 addresses this gap.

8.5. Future Work

Three directions remain beyond the items marked in this paper. First, closing the cross-layer refinement obligation, and extending the certificate model from Boolean to numeric observations, would extend the formal results to the composed system; deductive methods [44, 71] are a natural complement where bounded model checking does not scale. Second, replacing the fixed-arity aggregator with incremental recursion would make the recovery artifact constant-size for intervals of any length, and a live re-anchoring through an aggregate proof would extend the offline aggregation measurements to the full testbed. Third, replacing single-bridge DA trust with sampling across independent peers would let the DA sensor detect partial withholding rather than only outages.

9. Conclusion

A chain whose block validity requires settlement and DA receipts halts when those layers fail, although its validators are connected and correct: an external dependency can cause a liveness failure without a BFT failure. Engram makes suspending such a dependency an agreed validity and safety mode. A quorum-backed health certificate drives a four-state FSM above an unchanged CometBFT agreement core. In the degraded modes the corresponding receipt requirement is relaxed and cross-chain withdrawals are locked, and the return to ANCHORED requires recovery hysteresis, DA backfill of the SOVEREIGN interval, and a validator-verified recovery proof.

The evidence supports the mechanism within the stated assumptions. Bounded model checking found no violations in each layer (the largest run explored 4.54 million distinct states); the cross-layer refinement remains open. A seven-validator testbed with live Bitcoin and Celestia networks kept committing blocks under a composite fault and completed a full recovery cycle with complete AppHash agreement. Against seven vanilla validators on the same hosts, Engram lengthens the block interval by about 2% (1.262 s against 1.233 s).

Artifact Availability

The implementation, TLA+ specifications, and experiment scripts are in three repositories:

- `engram-sovereign-fsm`: the Cosmos SDK application, the TLA+ specifications, the recovery circuits, the experiment scripts, and this manuscript. `https://github.com/cuongct924/engram-sovereign-fsm`
- `engram-consensus-core`: the CometBFT fork used as the agreement core (Section 6.1 lists its four differences from upstream). `https://github.com/cuongct924/engram-consensus-core`
- `engram-barretenberg-wasm`: the build of the UltraHonk verifier embedded in the application; it is provenance only, not a build dependency. `https://github.com/cuongct924/engram-barretenberg-wasm`

The first two repositories are released under the Apache License 2.0. [PENDING: release tag or commit of each repository that produced the reported results.]

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Appendix

A. Extended Attack-Resilience Results

Table 21 reports the full ten-scenario single-actor attack suite. A1 and A2 ran on the earlier four-validator cluster (n=4); all other scenarios ran on the seven-validator testbed (n=7). The column *Byz. VP* gives the share of voting power held by the misbehaving validator: one of seven validators is 14.3%, and the attacker containers of A1 and A2 are non-validators (0%).

Table 21: Full Attack-Resilience Suite (live)

#	Attack	n	Byz. VP	Safety	Diverg.	Attack-specific signal
A1	Eclipse	4	0%	True	0	Filter holds 8/8, nil-prevote ratio 0.0 throughout
A2	Simulated multi-subnet Sybil	4	0%	True	0	Filter holds 8/8 despite 12 attackers
A3	Data withholding	7	14.3%	True	0	—
A4	Forged Bitcoin receipt	7	14.3%	True	0	—
A5	Withdrawal during Sovereign	7	—	True	0	Withdrawal correctly blocked; transaction never commits
A6	Malicious proposer	7	14.3%	True	0	Forged FSM state rejected by all honest validators (Section 7.1)
A7	Censorship	7	14.3%	True	0	Censoring confirmed real (target tx omitted 9/9 proposer turns); honest reject signals stayed at 0 since the next honest proposer included the tx before the ignore-round threshold fired
A8	Combined (A6 + A1)	7	14.3%	True	0	Safety held under simultaneous overlap
A9	Double-signing	7	14.3%	True	0	6/6 honest validators detected real duplicate-vote evidence; 1-block detection latency
A10	Timeout flood	7	14.3%	True	0	Cadence held (0.770→0.706 blocks/s); rate limiter dropped 0/0/0/0/0/0 at moderate (20 msg/s) intensity

A3 and A4 were instrumented only for safety and divergence, without an attack-specific metric such as A5's blocked-withdrawal flag or A9's detection latency.

At 25× its design threshold (500 msg/s vs. a 20 msg/s per-peer cap), the A10 timeout-flood attack still held cadence (0.825→0.789 blocks/s) while the rate limiter rejected the large majority of flood traffic (28,071 of ∼30,000 attempted messages on one validator); validator CPU usage roughly tripled (2.4–3.3%→7.85–9.51%, peak 22.40%) but stayed well short of saturation, and memory usage stayed flat. No attack in Table 21 drove block production toward zero, and resource usage under the P2P swarm attacks (A1/A2) stayed under 36% CPU, under 147 MiB memory per validator).

B. Backend Trade-off: Noir+Honk vs. Plonky3

This appendix compares the deployed Noir/UltraHonk recovery circuit (Section 4.7) with a STARK-based alternative, Plonky3. The comparison is necessarily partial: boundary-signature

verification, condition (e), dominates the deployed circuit's cost (1,097,274 opcodes with it versus 6,143 without, a factor of ∼180), and no Plonky3 configuration we built implements it.

Table 22: Deployed Circuit: Constant-Cost Scaling (Noir+Honk, Poseidon2)

Interval count	**Prove time**	**Verify time**	**Proof size**
1	38.3 s	25 ms	14,656 B
16	41.4 s	23 ms	14,656 B
256 ($N_{\max}$)	40.4 s	24 ms	14,656 B

Table 22 confirms that proving time, verification time, and proof size are independent of the number of real headers up to the compiled capacity $N_{\max}$=256. Verification by the validators' embedded WebAssembly verifier costs ∼26 ms per proof. An interval of L blocks requires $\lceil L/256 \rceil$ leaf proofs when submitted one by one, or a single aggregate proof up to 1024 blocks (Appendix C).

Table 23: Backend Comparison, Statement Coverage

Config.	(a)	(b)	(c)	(d)	(e)
Noir+Honk (deployed)	✓	✓	✓	✓	✓
Plonky3, primitive-only	–	–	–	–	–
Plonky3, fuller statement	✓	✓	✓	✓	–

(a) chain continuity, (b) FSM/withdrawal-lock policy legality, (c) circuit-breaker adherence, (d) state-root binding, (e) boundary ed25519-signature verification.

Table 23 makes the coverage gap explicit. Under a primitive-only comparison (256 chained Poseidon2 permutations, no FSM or signature logic, i.e. neither configuration implements condition (e)), Plonky3 produces a larger proof (1,278,939 B vs. 14,656 B, ∼87×) but proves faster (0.044 s vs. 0.684 s, ∼15.5×) with a transparent, post-quantum-secure setup versus the Noir/Honk backend's trusted KZG setup. Under a fuller-statement comparison that adds FSM legality and root binding (but still omits condition (e)), Plonky3 proves in 0.024–0.096 s with proof sizes of 84–124 KB across 2–256 headers. Neither comparison supports a general speed claim for Plonky3, because both omit the component that dominates the deployed circuit's cost. These numbers characterize primitive-level backend trade-offs only.

C. Chunk Aggregation

This appendix measures the aggregator of Section 4.7 against submitting the chunk proofs one by one. The measurements were taken offline on a 10-core, 32 GiB macOS workstation, not on Testbed A, with nargo 1.0.0-beta.22 and `bb` 5.0.0-nightly.20260522. The chain is synthetic: it has one deterministic header sequence with real ed25519 precommits and one validator set, so that consecutive chunks chain through their roots and heights. Aggregate proving and verification are averaged over five trials per interval length; each leaf proof was timed once. Absolute times are not comparable to Table 22, which was measured on a different host.

The aggregator verifies the chunk proofs in-circuit and checks that chunk i starts at the root and height where chunk $i-1$ ended, under the same validator-set hash. It exposes the same public inputs as a single chunk, so an aggregate proof stands in for a leaf proof, and its capacity K=4 is fixed at compile time. Unbounded incremental recursion would require the step circuit to receive its own verification-key hash as a public input, which we did not implement. Recursion is applied at the chunk level because in-circuit verification of one UltraHonk proof costs about 0.7 million gates, whereas hashing one header costs about 12 opcodes: recursion over per-header proofs measured 650–1,200× slower than direct hashing at matched sizes.

Table 24: Chunk Aggregation vs. Submitting Chunk Proofs One by One (K=4)

	One by one			**Aggregate**		
L	Prove	Bytes	Verify	Prove	Bytes	Verify
256	31.5 s	14,656	19.8 ms	57.2 s	14,656	20.2 ms
512	60.5 s	29,312	39.5 ms	89.9 s	14,656	20.2 ms
1024	120.1 s	58,624	79.1 ms	146.7 s	14,656	20.2 ms

Verify is the validators' embedded WebAssembly verifier, measured per proof (30 iterations, three runs); the one-by-one column is n times the leaf cost. Aggregate prove includes the leaf proofs. At L=256 a single leaf proof is submitted and the aggregate row is shown for the overhead only.

Table 24 shows that submitted bytes and verification time of the aggregate do not depend on L up to $K \cdot 256 = 1024$, while the one-by-one cost grows linearly. The price is one extra proof: the aggregator circuit has 3,002,299 gates and proves in 25.7–29.5 s using about 5.2 GB, against 3,234,226 gates, 30.0–31.5 s, and 5.7–5.9 GB for a leaf. Total proving time rises by 49% at L=512 and 22% at L=1024. Chunk proofs were produced sequentially; proving them in parallel was not measured. Every chunk boundary height needs a round-zero commit, so a round skip at an interior boundary prevents building that chunk's witness.

The aggregator rejects each of eleven malformed witnesses: reversed or repeated chunks, a wrong total count, final root, first height, or validator-set hash, a height gap between chunks, a different validator set in the second chunk, an extension through padding slots, and a chunk count outside $[1, K]$. The valid two-chunk witness is accepted, and a 512-header aggregate passes every `SubmitRecoveryProof` binding against fixture chain state. A live re-anchoring through an aggregate proof on the Docker testbed was not run.

D. Bitcoin Reorg Depth Reaction

Section 5.4 states a threat-model boundary – safety is modeled given Bitcoin reorganizations no deeper than a configurable finality depth $K_{\text{DEEP_FINALITY}}$ – as a precondition of the formal result. This appendix tests the implementation at and beyond that boundary with forced Bitcoin reorganizations, each confirmed from the resulting block hashes, at five depths in one seven-validator run: {1, 2, 3, 5, 15}, with $K_{\text{DEEP_FINALITY}}$=2 and a suspicious-state threshold of 5.

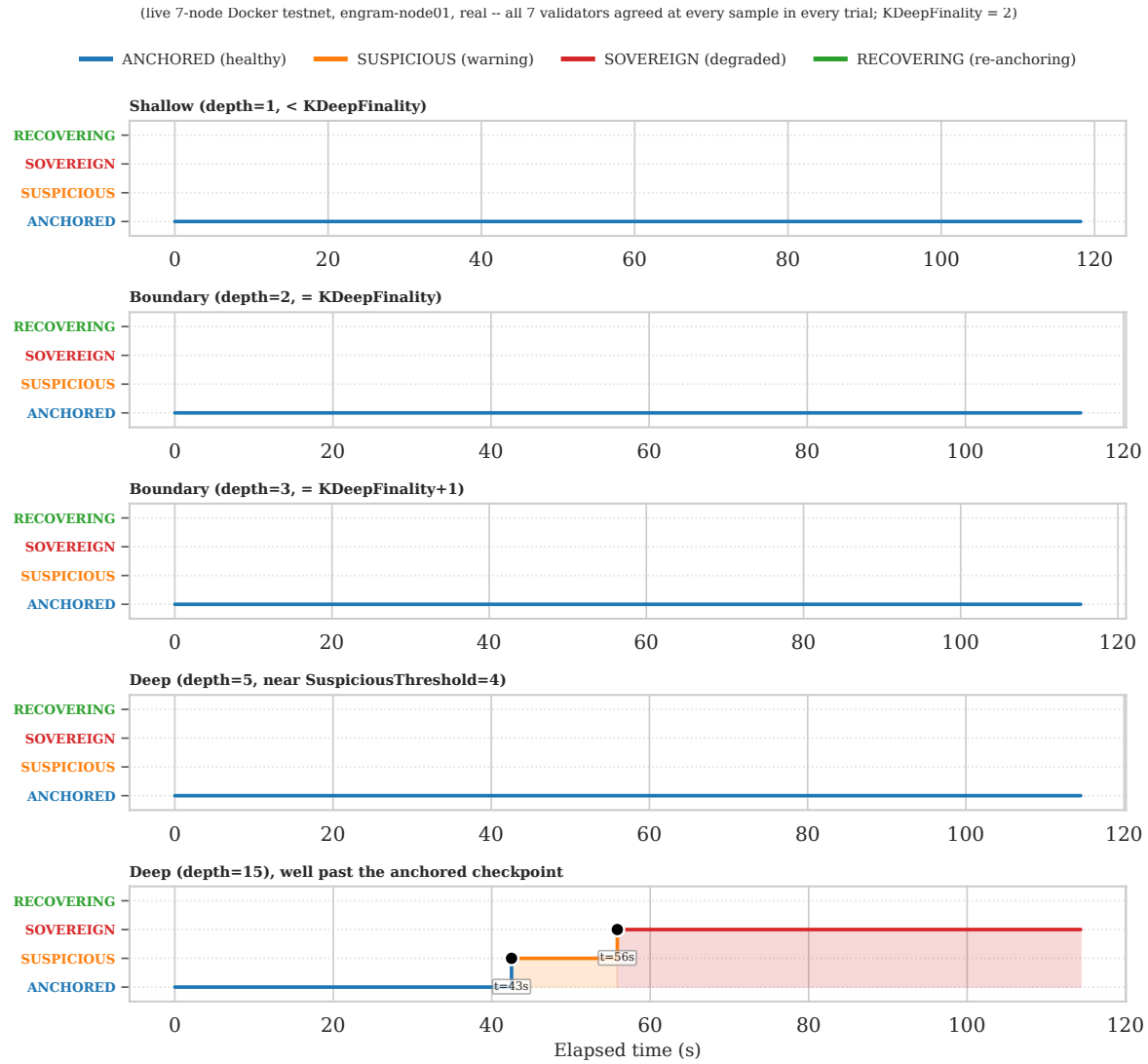


Figure 8: FSM reaction to Bitcoin reorgs at five depths, one trial per depth, seven-validator testbed. All 7 validators agree at every one of 181 polled rounds per trial (1,267 samples total, 0 RPC errors).

Table 25 summarizes the result. Depths 1–5 all orphan real, independently-confirmed blocks, yet none of them trip the FSM within the observation window: all 7 validators stay ANCHORED throughout. At depth 15, all 7 validators transition ANCHORED→SUSPICIOUS at t=43 s and →SOVEREIGN at t=56 s, in lockstep. This path rules out a direct critical-condition jump: the FSM's ANCHORED branch bypasses SUSPICIOUS entirely when the committed certificate's anchor-reverification flag is set, but the observed path instead passes through SUSPICIOUS, and the suspicious-duration counter reached 23 of the 24-block escalation bound (MaxSuspiciousTime) in the last sample before the SOVEREIGN transition. The depth-15 reaction is attributable to this duration-based escalation, not to anchor re-verification failing.

The crossover between depth 5 (no reaction) and depth 15 (reaction) was not swept at finer granularity. Whether anchor re-verification ever set the anchor-reverification flag during this run is not established by the available logs – the per-validator sensor value was not captured, only the committed FSM state and suspicious-duration counter. The data that is available shows the depth-15 escalation came from the suspicious-duration bound, not from a direct critical-condition jump.

Table 25: FSM Reaction to Real Bitcoin Reorgs by Depth

Depth	Reorg confirmed?	FSM reaction
1 (below K_{DEEP})	N/A (too shallow)	Stays ANCHORED
2 (= K_{DEEP})	Yes	Stays ANCHORED
3 (= K_{DEEP}+1)	Yes	Stays ANCHORED
5	Yes	Stays ANCHORED (within window)
15 (past checkpoint)	Yes	SUSPICIOUS (43s) → SOVEREIGN (56s)

E. Network Emulation Parameters

Table 26 lists every `tc netem` profile, applied through Pumba. No correlation or distribution option is set, so delay jitter is uniform and loss is independent per packet (the `netem` defaults). Profiles that name no interface use Pumba's default interface (`eth0`), which carries Bitcoin, Celestia, and RPC traffic rather than the dedicated pairwise validator links. Only the S4 scenario and the relay-latency attack (P4) of Section 7 target the validator links explicitly.

Table 26: Network Emulation Profiles (Testbed A)

Profile	Effect and target	Duration	Used by
`chaos-delay`	100 ms delay, 20 ms jitter, all seven validators	5 min	baseline
`chaos-loss`	5% loss, validators 1 and 2	2 min	composite
`chaos-eclipse`	100% loss, validator 1	3 min	S5
`chaos-btc-delay`	500 ms delay, 100 ms jitter, one `bitcoind`	2 min	composite
`chaos-wan-latency`	Per-validator delay and jitter, Table 27	10 min	WAN baseline
`chaos-wan-loss`	Per-validator loss, Table 27	10 min	WAN variant
S4 eclipse	300 ms delay, 50 ms jitter on the six validator links of validator 1	2 min	S4
P4 relay latency	350 ms delay, 50 ms jitter on all P2P interfaces of one validator	244 s	P4

Table 27: WAN Baseline Profiles Per Validator (one profile of seven emulated regions)

Validator	1	2	3	4	5	6	7
Delay (ms)	15	70	140	45	100	30	180
Jitter (ms)	3	15	25	10	20	6	30
Loss (%, loss variant)	1	3	6	2	4	1	5

Each validator receives one delay value on its egress, so the profile is a per-validator offset rather than a pairwise round-trip-time matrix. A pairwise matrix between regions is not measured on Testbed A.